

Introduction

Modelling and understanding how a car behaves is the daily mission of every vehicle dynamicist. In this context, predicting the vehicle's response to driver-controlled inputs (such as steering actions) or external disturbances (such as road bumps) lies at the core of vehicle dynamics studies.

This report examines both lateral and vertical dynamics through step-response analyses applied to simplified models.

The first chapter focuses on the bicycle model, where a steering step input highlights the buildup of yaw motion for different Nissan GT-R configurations, while the second one turns to vertical dynamics, aiming to optimize (with different level of complexity) SDOF and TDOF quarter car models of a Ford Granada subjected to a bump excitation.

Simulations are performed in ADAMS, with results subsequently validated using MATLAB-Simulink to ensure accuracy and consistency across different simulation environments. Together, these analyses demonstrate how step-response studies, combined with carefully chosen models and cross-validated simulations, provide a coherent understanding of vehicle dynamics in multiple domains.

1 Bicycle model and handling analysis

The relevance of bicycle model in automotive industry lies in its ability to reduce the system's complexity by approximating the vehicle as a single-track model, limiting the number of degrees of freedom: two translations (longitudinal and lateral) and one rotation (yaw) are allowed. Even though pitch, roll, and heave motions are neglected, the model retains the essential dynamic characteristics required to accurately describe planar vehicle behaviour.

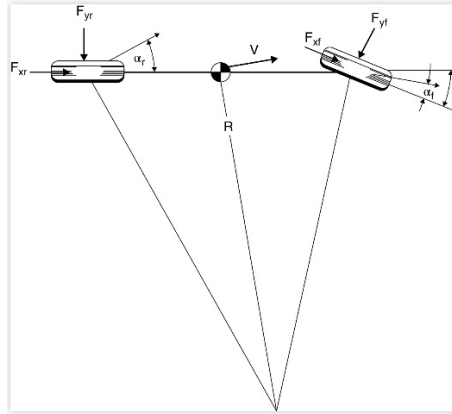


Figure 1: Bicycle model (Gillespie, 2021)

Among different types of analysis that can be carried out with the bicycle model, the steady-state response of the system is going to be investigated when a step steering input is applied while traveling with constant longitudinal speed. Under these conditions, first equation of motion can be neglected if also yaw rate r and lateral velocity v are assumed to be small, leading to the following equations:

$$m(\dot{v} + rU) = F_{yf} + F_{yr} \quad (1)$$

$$I\dot{r} = aF_{yf} - bF_{yr} \quad (2)$$

The lateral forces from Eq.(1) and Eq.(2) can be expressed in terms of cornering stiffnesses and tire slip angles, which are defined as the angles between the wheel direction of heading and its plane: they can be further related to the state variables (v, r), adopting small-angles approximation.

$$F_{y_f} = -C_f \cdot \alpha_f \quad (3)$$

$$F_{y_r} = -C_r \cdot \alpha_r \quad (4)$$

$$\alpha_f = -\delta + \arctan\left(\frac{v_{w,f}}{u_{w,f}}\right) \cong -\delta + \frac{v_{w,r}}{u_{w,r}} = -\delta + \frac{v+ar}{U} \quad (5)$$

$$\alpha_r = \arctan\left(\frac{v_{w,r}}{u_{w,r}}\right) \cong \frac{v_{w,r}}{u_{w,r}} = \frac{v-br}{U} \quad (6)$$

Substituting earlier expressions into the equations of motion, the derivatives of the states ($\dot{r}, \dot{v}$) can be linearly linked to the states (r, v) and the input (δ).

$$m(\dot{v} + rU) = -\frac{(C_f+C_r)}{U}v - \frac{(aC_f-bC_r)}{U}r + C_f\delta \quad (7)$$

$$I\dot{r} = -\frac{(aC_f-bC_r)}{U}v - \frac{(a^2C_f+b^2C_r)}{U}r + aC_f\delta \quad (8)$$

For the development of the bicycle model, *Nissan GT-R* was chosen as a target vehicle with real parameters listed in Appendix A.

1.1 Linear tyre model

In a linear tyre model, the cornering stiffness values for each vehicle configuration are assumed constant, accounting for the effect of the vertical load F_z on each axle (as detailed in Table IV, Appendix A).

	Mass (kg)	Load distribution	a (mm)	b (mm)	F_{z_f} (N)	F_{z_r} (N)	C_f (N/deg)	C_r (N/deg)
US	1739	54%f - 46%r	1278.8	1501.2	9212	7847	1099	1717
NS	1739	39%f - 61%r	1695.8	1084.2	6653	10406	1072	1686
OS	1739	30%f - 70%r	1946	834	5118	11942	977	1625

Table 1.1: Vertical loads and cornering stiffnesses evaluation

Consistent with the design of most passenger vehicles, the Nissan GT-R Nismo base configuration exhibits understeering behaviour, which can be quantitatively demonstrated by evaluating its understeer gradient.

$$k_{us} = \frac{m}{l} \cdot \left(\frac{b}{C_f} - \frac{a}{C_r} \right) \quad (9)$$

Acting on load distribution, the position of the centre of gravity (and tyre cornering stiffness consequently, depending on F_z) was adjusted to obtain neutral-steering and oversteering behaviours from Eq.(9).

	k_{us} ($rad/\frac{m}{s^2}$)	
US	0.0068	As speed increases, curvature response becomes smaller requiring more steering input
NS	6.3E-05	The corner radius is nearly independent from speed: it can be estimated through the Ackerman steering geometry
OS	-0.0038	As speed increases, curvature and yaw rate response tend to infinity

Table 1.2: Understeer gradient evaluation

The behaviour of each configuration can be analysed through the curvature gain from Eq.(10), with results shown in Fig.2.

$$\frac{1}{R\delta} = \frac{1}{l+k_{us}\cdot U^2} \quad (10)$$

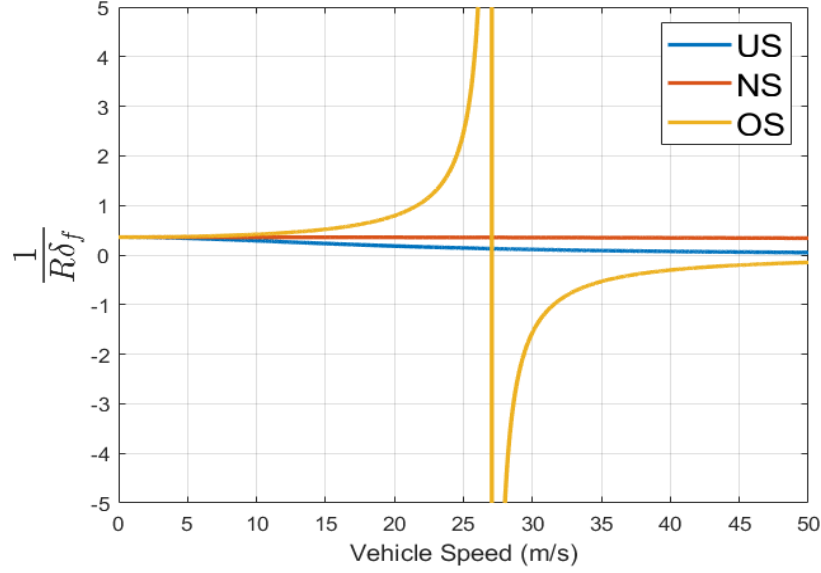


Figure 2: Curvature gain for different Nissan GT-R configurations

On this basis, the required speed for testing the model is fixed at $U = 15\text{m/s}$ to prevent the oversteering model from reaching instability at critical velocity and to enhance differences between the models when they start to behave differently.

The models were developed on ADAMS, using three rigid body to represent front tyre, rear tyre and vehicle body, and applying three different joints: fixed joint between rear tyre and body, revolute joint between front tyre and body and planar joint between the body and the ground.

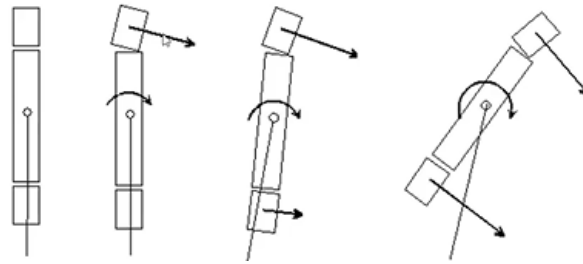


Figure 3: Transient cornering effect (Gillespie,2021).

As shown in Fig.3, when a step steering input ($\delta = -15\text{deg}$) is applied at the front wheel, it produces a front slip angle and the corresponding tire lateral force. This force creates a yawing moment around the vehicle's centre of gravity, generating a yaw acceleration. As the vehicle begins to build yaw rate, the rear tire also develops a slip angle and, consequently, a lateral force. The rear lateral force counteracts the front one, reducing the yaw acceleration until the yaw rate is stabilized (Gillespie, 2021).

This cornering mechanism can be visualized from Table 2.1 and 2.2, comparing the different configurations.

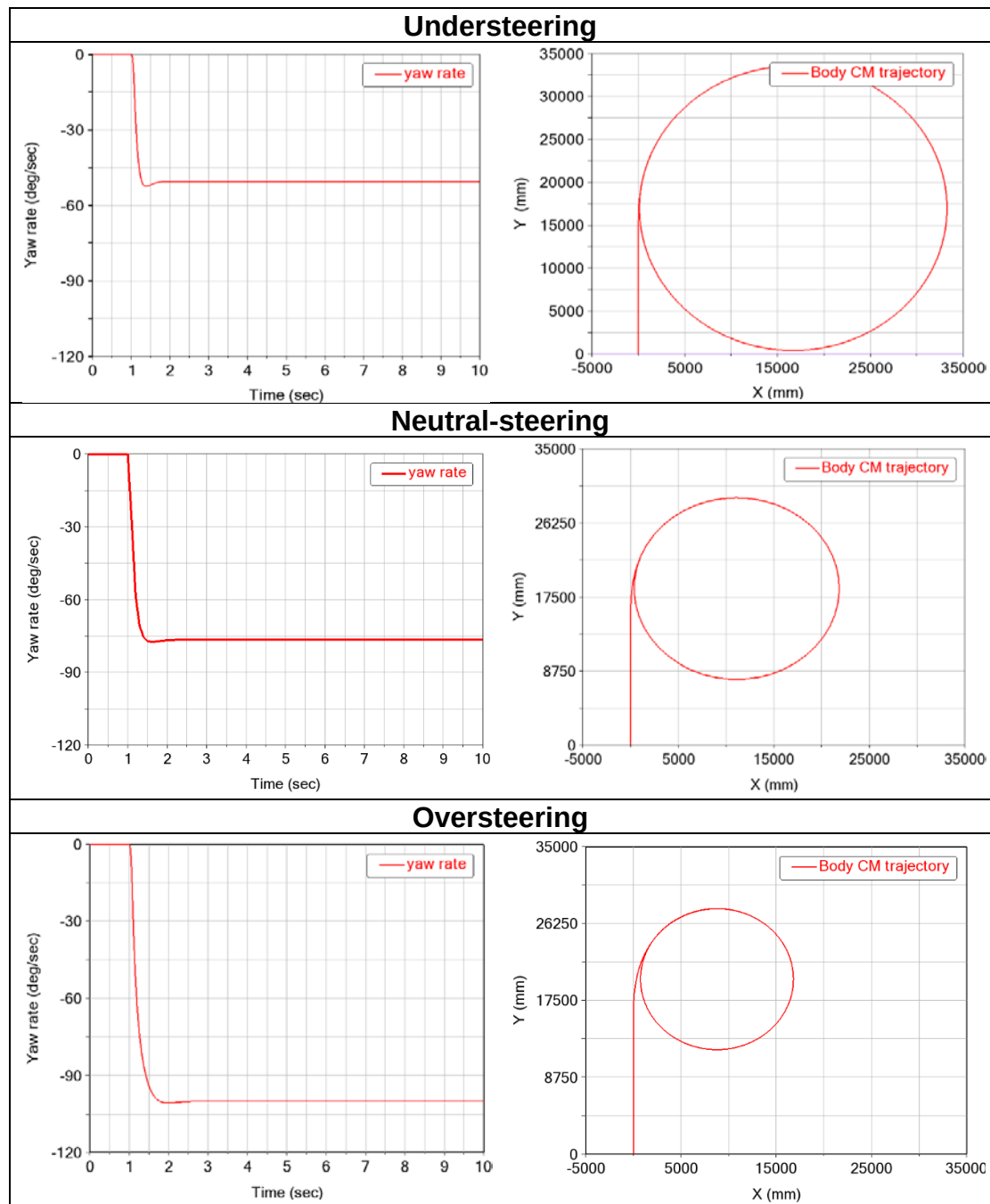


Table 2.1: Linear models comparison on ADAMS

Each vehicle configuration generates a different yaw rate response for the same steering input. As the rear axle contributes to a larger portion of the lateral force (going from US to OS), the expected steady-state yaw rate increases. The vehicle's trajectory on a global reference frame shows the vehicle deviating outward (US), aligning more closely with the intended path (NS), or rotating more sharply than expected (OS) depending on the configuration.

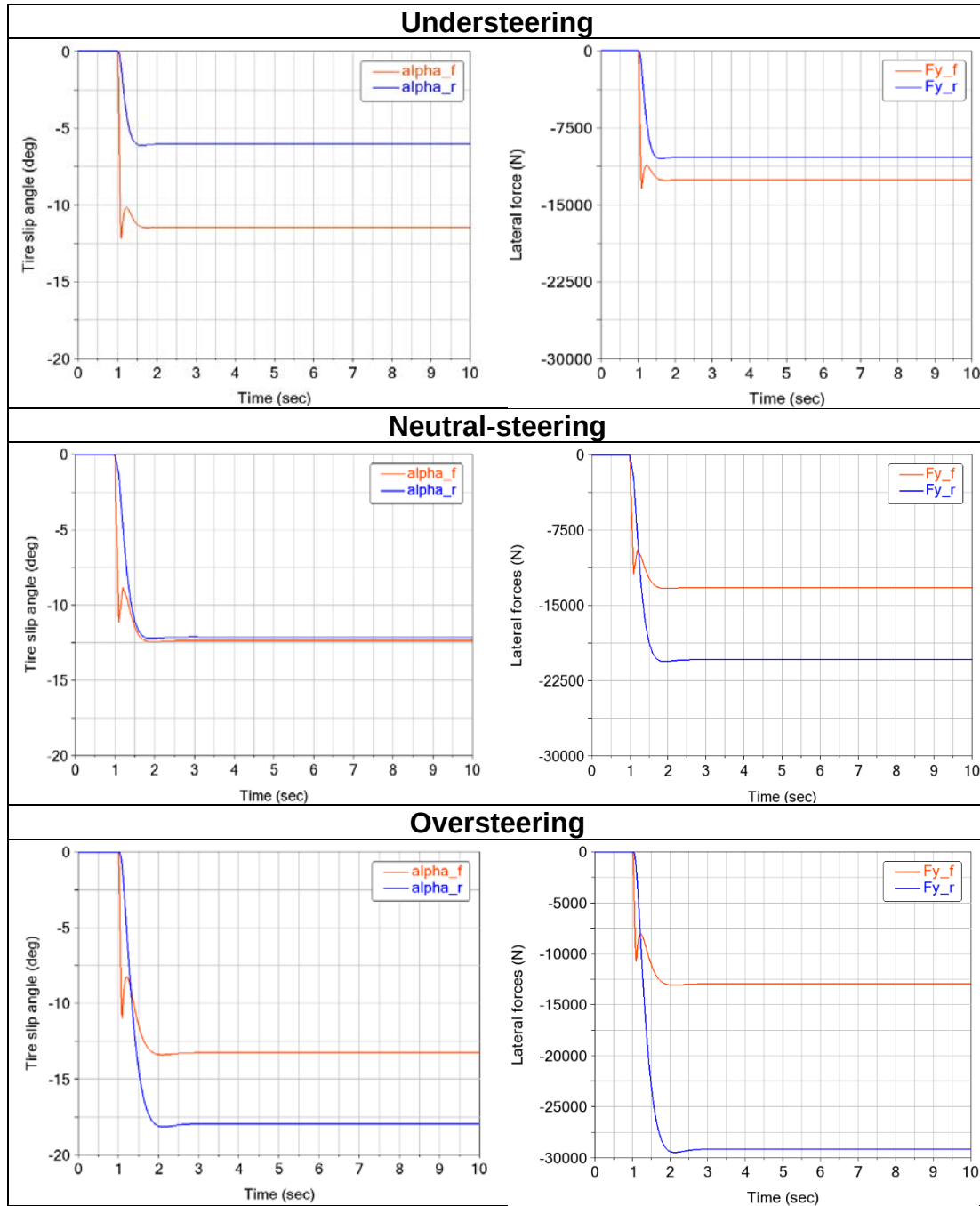


Table 2.2 : Linear models comparison on ADAMS

In an ideally balanced neutral-steering configuration, slip angle at the front tire is equal to the rear one and vehicle's trajectory follows the expected geometrical curvature with minimal deviation.

According to Eq.(2) and Eq.(3), it must be accounted that negative slip angles produce positive lateral forces but for sake of simplicity the analysed forces are taken negative, being obtained as product of a state variable ($\alpha < 0$) and a design variable ($C > 0$) in ADAMS.

1.2 Linear comparative analysis on MATLAB-Simulink

The state equations can be further developed into a matrix form widely used in the literature. (Ren et al., 2014)

$$\begin{bmatrix} \dot{v} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} -\frac{C_f + C_r}{mu} & -u - \frac{aC_f - bC_r}{mu} \\ -\frac{aC_f - bC_r}{I_z u} & -\frac{a^2 C_f + b^2 C_r}{I_z u} \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} + \begin{bmatrix} \frac{C_f}{m} \\ \frac{aC_f}{I_z} \end{bmatrix} \delta_f \quad (11)$$

The state space matrices shown in Eq.(11) are implemented in MATLAB-Simulink through the state space block shown in Fig.4, with state variables taken as output if matrices C and D are properly set (as detailed in Appendix E).

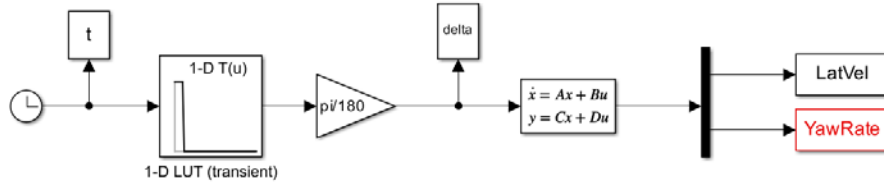


Figure 4: State space implementation

Through a 1-D Lookup table (LUT) that accounts for the transient of the step steering, the model above allows to perform comparative analysis with results carried out in ADAMS from previous section, including from Eq.(3) up to Eq.(6) inside the MATLAB function blocks shown in Fig.5.

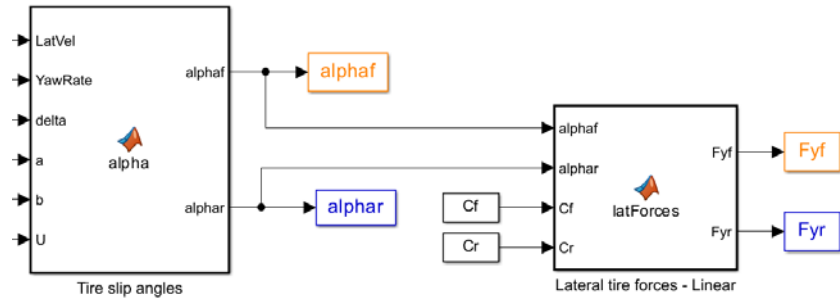
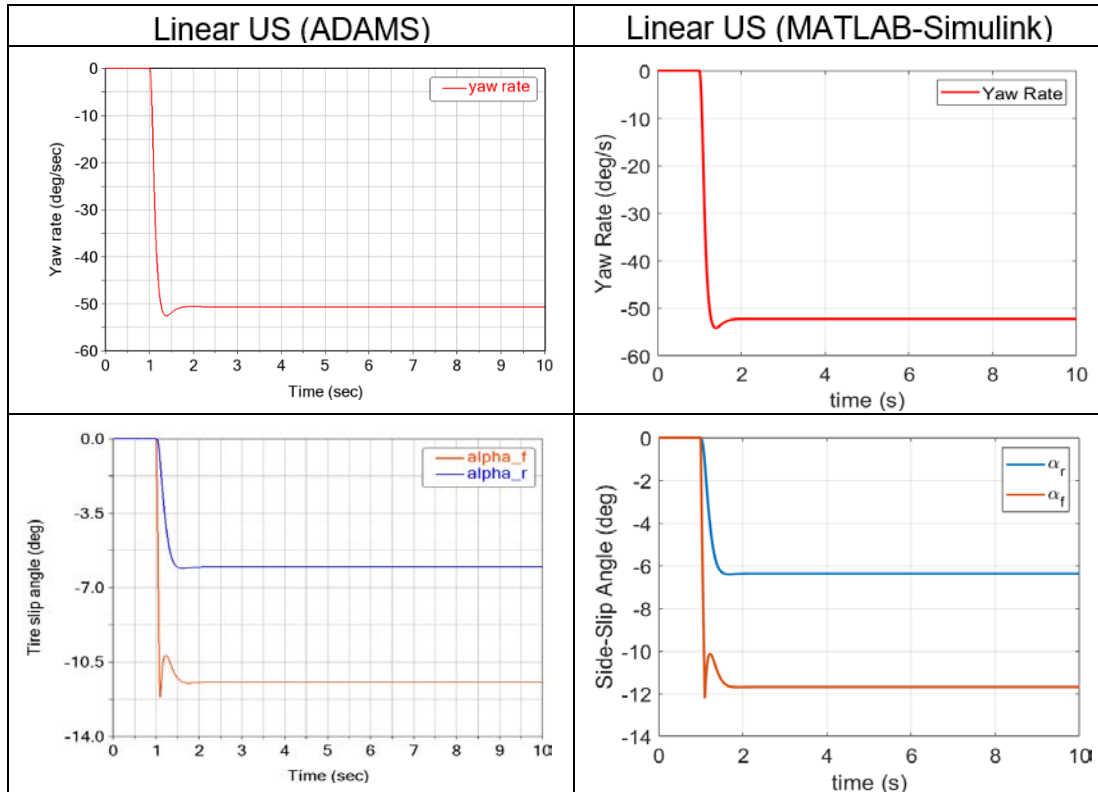


Figure 5: Slip angles and lateral forces implementation



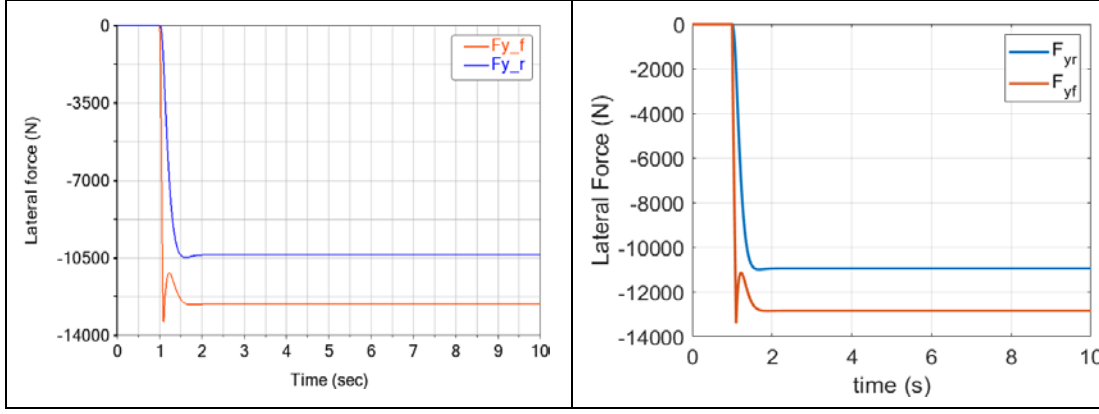


Table 3: Linear model comparison

Comparing yaw rate, tire slip angles and lateral forces (with numerical results listed in Appendix B), it can be observed from Table 3 how the state space model developed on Simulink closely matches the results from ADAMS, being also able to capture the transient with high accuracy.

1.3 Derivative analysis

Analysis of the derivatives can be performed on MATLAB to evaluate the step steer response of the understeering configuration.

The derivatives of stability are estimated from the equation of motion, as shown in Fig.6.

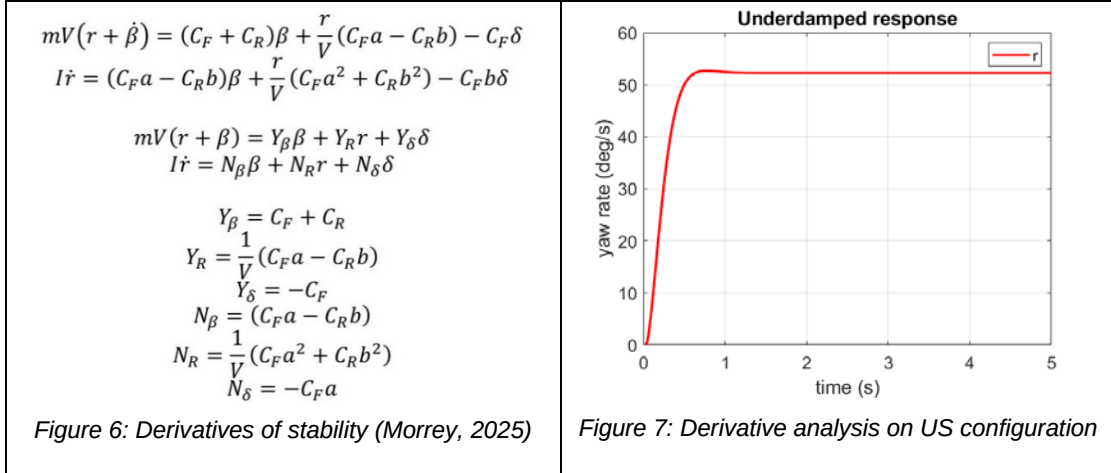


Figure 6: Derivatives of stability (Morrey, 2025)

Figure 7: Derivative analysis on US configuration

If steering input and longitudinal velocity of previous case studies are assumed, the computation leads to a value of $\zeta = 0.841$ (as reported in Appendix B results table), defining an underdamped response whose steady state value is coincident with the output of the linear state space but with opposite sign (52.25 deg/s), as it is shown in Fig.7. The main difference concerns a smaller overshoot compared to the simulation of the US model from Table 3.

1.4 Non-linear tyre model

As slip angle increases due to large steering maneuvers, the predicted responses from the linear bicycle model are significantly different compared to those of actual vehicles. This is due to the nonlinear relationship between the tire lateral force and tire slip angle, as shown in Table 4. (Ren et al., 2014)

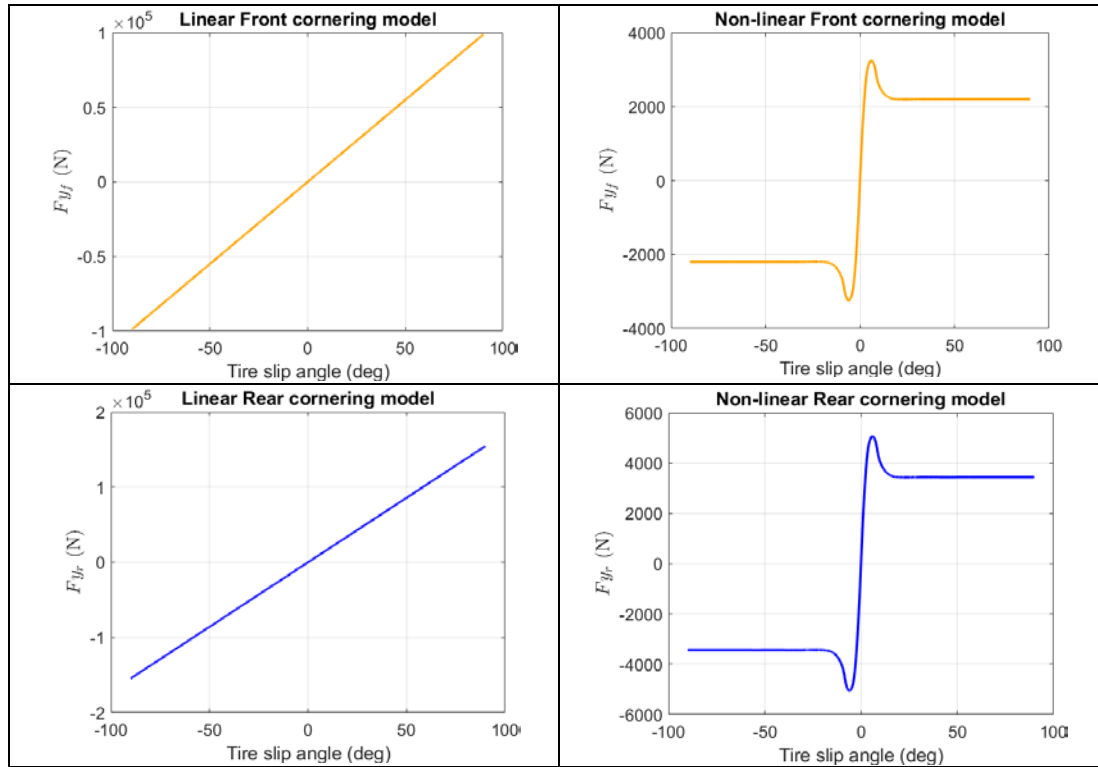


Table 4: Linear vs non-linear cornering models

These non-linear cornering models are scaled using the linear values of C_f and C_r (as detailed in Appendix E) and are implemented as splines, allowing a direct comparison between the linear and non-linear models.

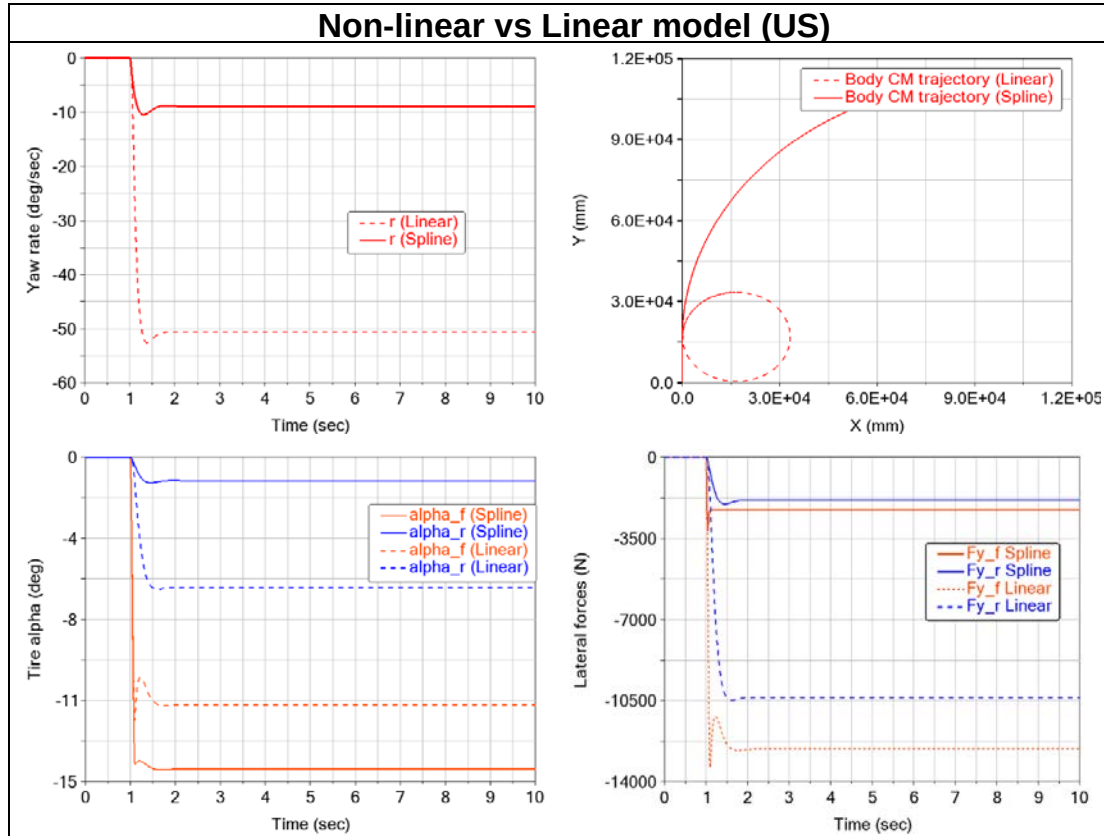


Table 5: Comparison between linear and non-linear models on ADAMS

Table 5 illustrates how the non-linear model captures the reduction in cornering stiffness as the tire approaches its friction limits. Consequently, it predicts lower yaw rates and lateral forces, providing a more realistic representation of vehicle behaviour during aggressive maneuvers since the linear bicycle model is unable to account for tire force saturation.

1.5 Non-linear comparative analysis on MATLAB-Simulink

For a non-linear bicycle model, it is no longer possible to define linear state space matrices as it was done in section 1.2.

Without any changes on the input side, the new model incorporating the spline for both rear and front tire forces is implemented in Simulink, accounting for the model equations shown below. (Novara, 2025)

$$\dot{X} = v_x \cos(\psi) - v_y \sin(\psi) \quad (12)$$

$$\dot{Y} = v_x \sin(\psi) + v_y \cos(\psi) \quad (13)$$

$$\dot{v}_x = v_y \omega_\psi + a_x \quad (14)$$

$$\dot{v}_y = -v_x \omega_\psi + \frac{1}{m} (F_{yf} + F_{yr}) \quad (15)$$

$$\dot{\psi} = \omega_\psi \quad (16)$$

$$\dot{\omega}_\psi = \frac{1}{J} (a F_{yf} - b F_{yr}) \quad (17)$$

Performing the integration of the outputs, the yaw rate can be obtained (coincident with ω_ψ). Tire slip angles are evaluated implementing Eq.(5) and Eq.(6) inside the “*Tire slip angles*” block as done for the linear model, while LUTs replace the linear relations from Eq.(3) and Eq.(4), as shown in Fig.8.

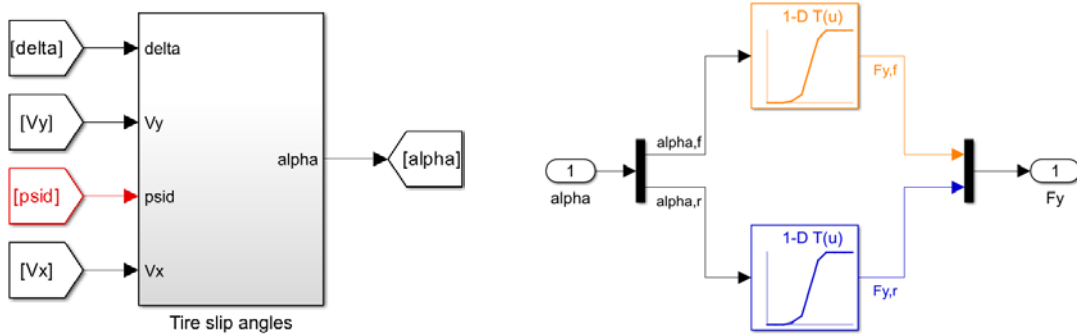
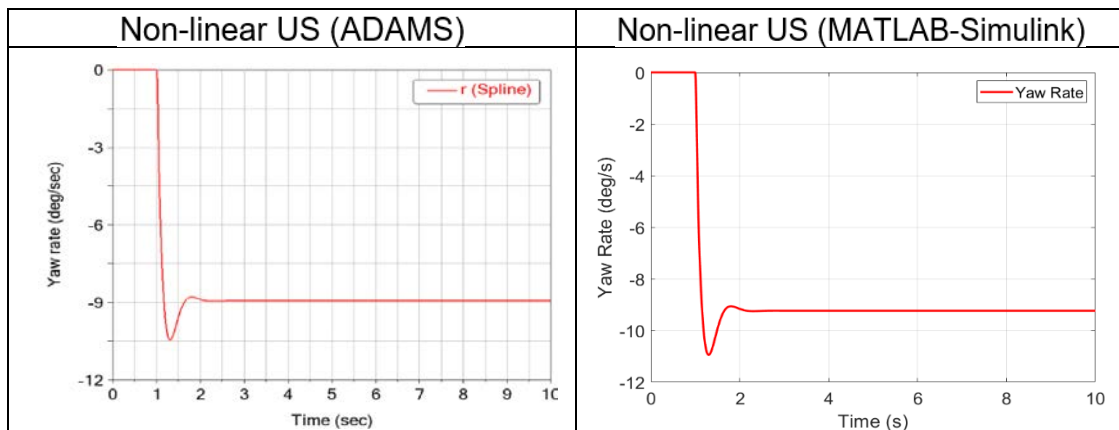


Figure 8: Tire slip angles and non-linear lateral forces evaluation on Simulink



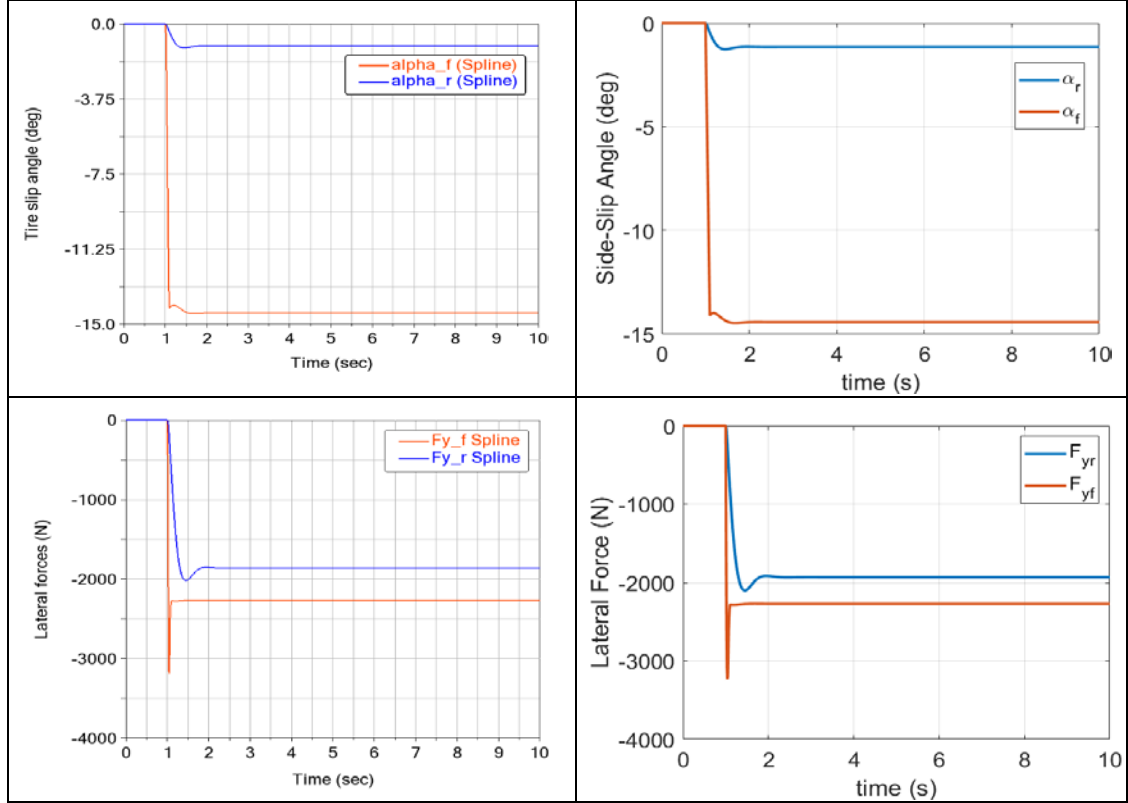


Table 6: Non-linear model comparison

As previously stated for the linear model, also the non-linear model shows that yaw rate, tire slip angles, and lateral forces obtained in Simulink closely match the results from ADAMS, with numerical values reported in Appendix B.

1.6 Technical conclusions

Previous sections have demonstrated that the bicycle model can provide a quantitative measure of the vehicle's expected handling, facilitating the identification of critical speed at which oversteer instabilities may occur through the estimation of the understeer gradient, thereby supporting proactive design and safety considerations.

Furthermore, the model enables prediction of yaw rate responses and trajectory paths, offering insights into vehicle stability and control before any driving test. Beyond its predictive capabilities, it also serves as a diagnostic tool by helping to isolate the causes of observed deviations in vehicle behavior, such as aerodynamical moments in real driving conditions.

Although the model provides a clear framework that bridges theoretical analysis and practical testing, it is not without limitations: for instance, effects of camber, pitch and roll were not considered in this analysis. Models with a higher number of degrees of freedom, such as the double-track model, are required to accurately estimate the lateral weight transfer during cornering.

Additionally, as it was mentioned in section 1.4, some limitations concern the tire lateral force characteristics to slip angle: it can be approximated as a linear relationship only when a vehicle performs a lateral maneuver with less than 0.3 g of lateral acceleration (Ren et al., 2014). To account for large lateral acceleration, it is necessary to implement a tire model that can reflect the non-linear tire behaviour.

2 Suspension dynamics and ride optimisation

In the design of modern suspension systems, engineering priorities inevitably converge around two fundamental requirements: performance and comfort. From a performance standpoint, due to the tire lateral force dependency on the vertical load, the minimization of the contact patch load variation is desirable. From a comfort perspective, the suspension system is expected to reduce the vertical acceleration of the chassis: a reduced magnitude of the acceleration felt by an occupant results in a higher perception of comfort.

Balancing these two objectives leads to the implementation of simplified quarter car models, such as single degree of freedom (SDOF) and two degrees of freedom (TDOF).

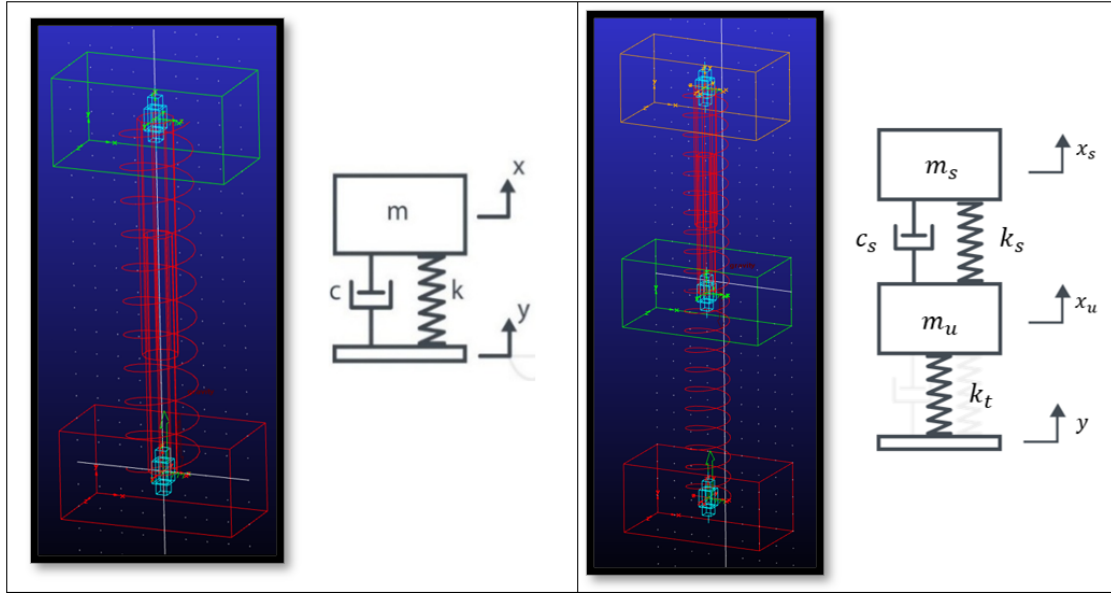


Figure 9: SDOF and TDOF models

In particular, the aim of optimizing the variation in contact patch force is going to be pursued at different levels: starting from the simplest model (SDOF) with linear damping, the level of complexity is increased performing the Design of Experiments procedure also on a TDOF, with a non-linear damper curve incorporated inside the model. To evaluate the system response, a step input with 25mm amplitude is used to model a road bump.

For the development of these models, Ford Granada was chosen as a target vehicle, with lumped parameters listed in Appendix C.

2.1 SDOF model

In a SDOF model, the quarter car mass is assumed to be connected to the road through a spring-damper element. The model was developed on ADAMS, introducing translational joints for each body (road and vehicle mass) to admit the vertical displacement. Its behaviour is described from the equation of motion:

$$m\ddot{x} + c(\dot{x} - \dot{y}) + k(x - y) = 0 \quad (18)$$

Simulating a bump through a step input, the model allows to catch the trend of vertical acceleration and the load variation at contact patch, with validation performed through the implementation of Eq.(18) on Simulink as shown below:

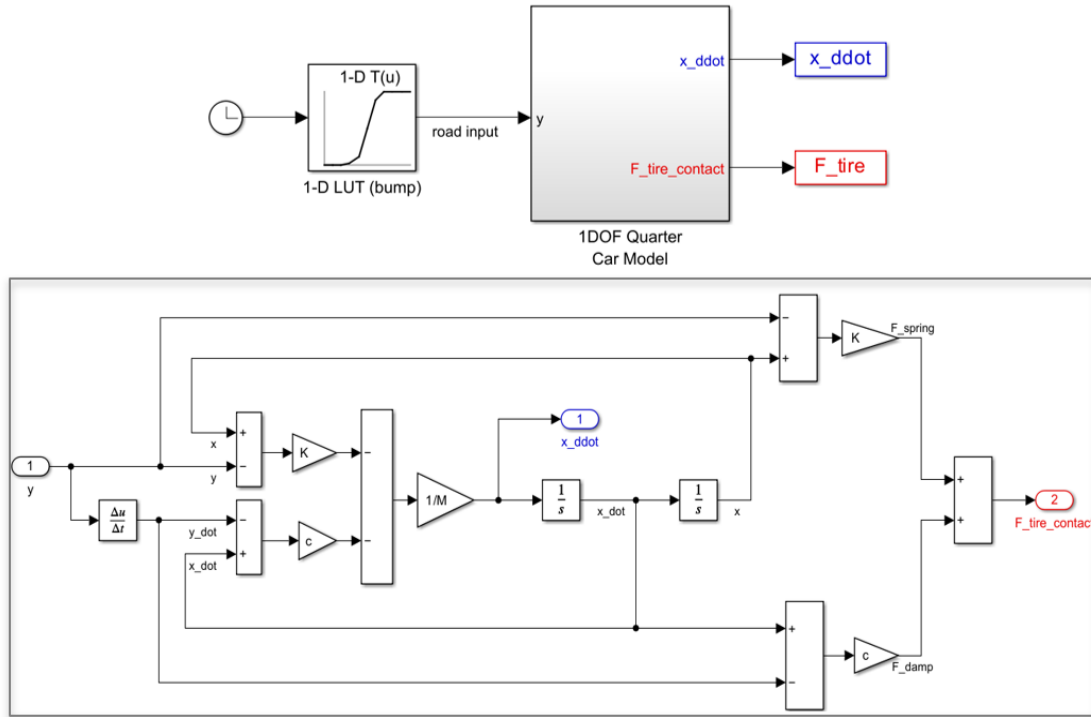


Figure 10: SDOF model implementation

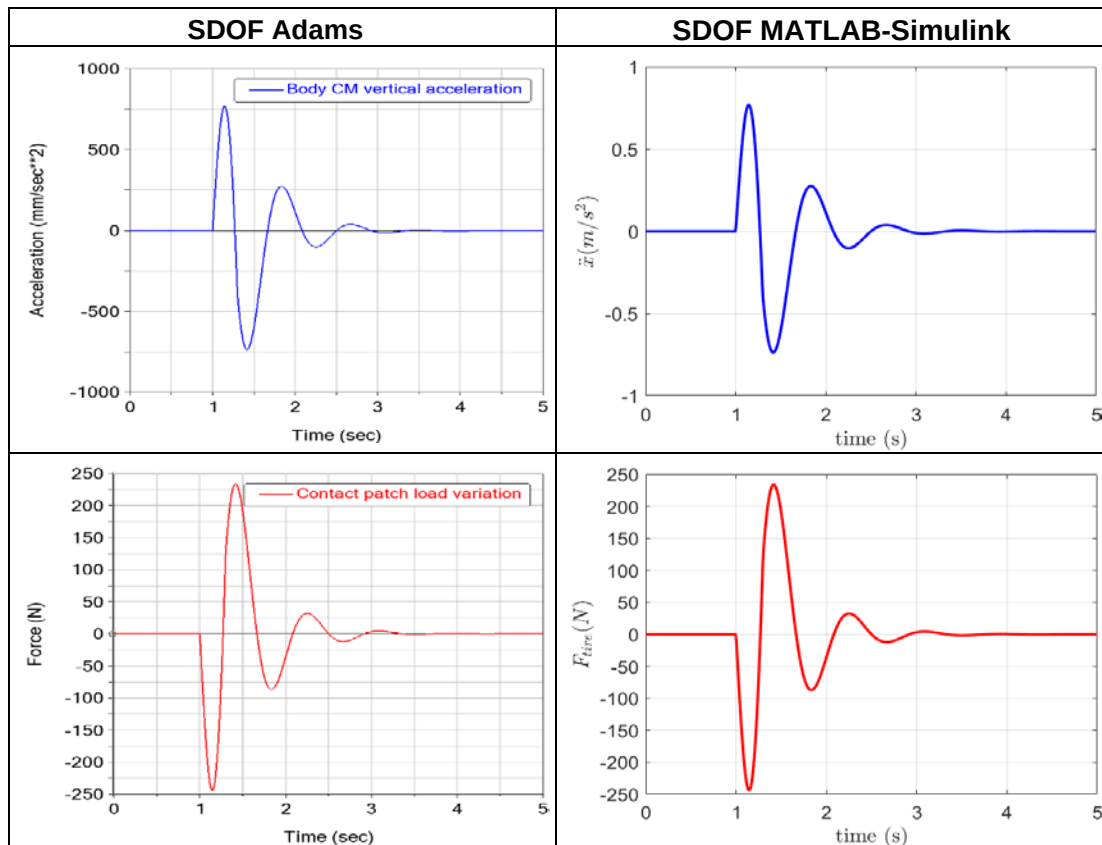


Table 8: SDOF results comparison (linear damping)

2.2 TDOF model

The TDOF model accounts for the distinction between sprung and unsprung masses. The model was developed on ADAMS, introducing translational joints for each body created (road, unsprung mass and sprung mass) to allow the vertical displacement. Its behaviour is described from second order differential equations:

$$m_s \ddot{x}_s + c_s (\dot{x}_s - \dot{x}_u) + k_s (x_s - x_u) = 0 \quad (20)$$

$$m_u \ddot{x}_u + c_s (\dot{x}_u - \dot{x}_s) + k_t (x_u - y) = 0 \quad (21)$$

The previous set of equations can be organized in state space form on MATLAB (as detailed in Appendix E), using the command *bode()* to show how the system behaves at different frequencies, as an alternative to the manual implementation of a sine-sweep road input in ADAMS.

The analysis was carried out on $\ddot{x}_s$ and F_{tire} by varying the damping ratio, with two magnitude peaks corresponding to the natural frequencies of the system as shown in Fig.11.

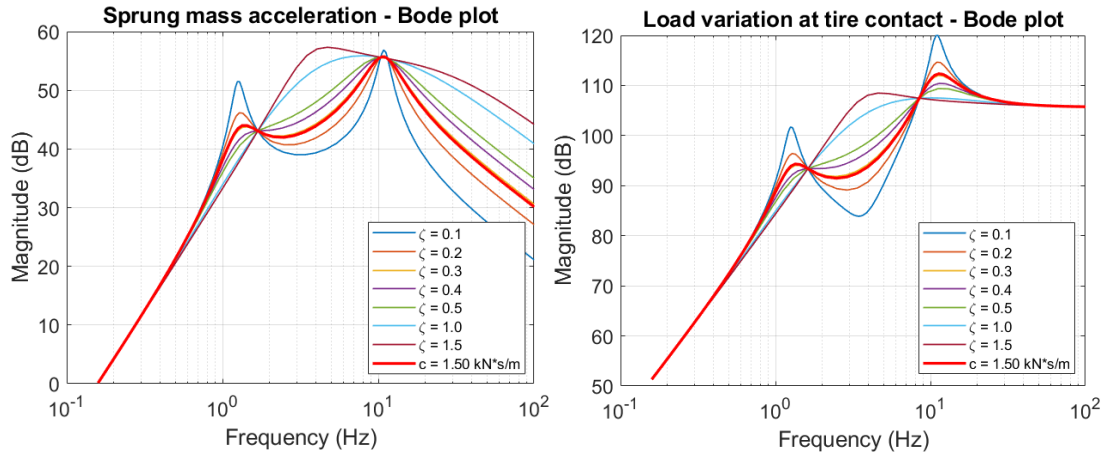
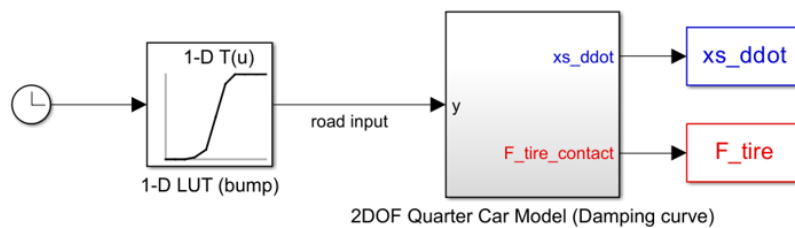
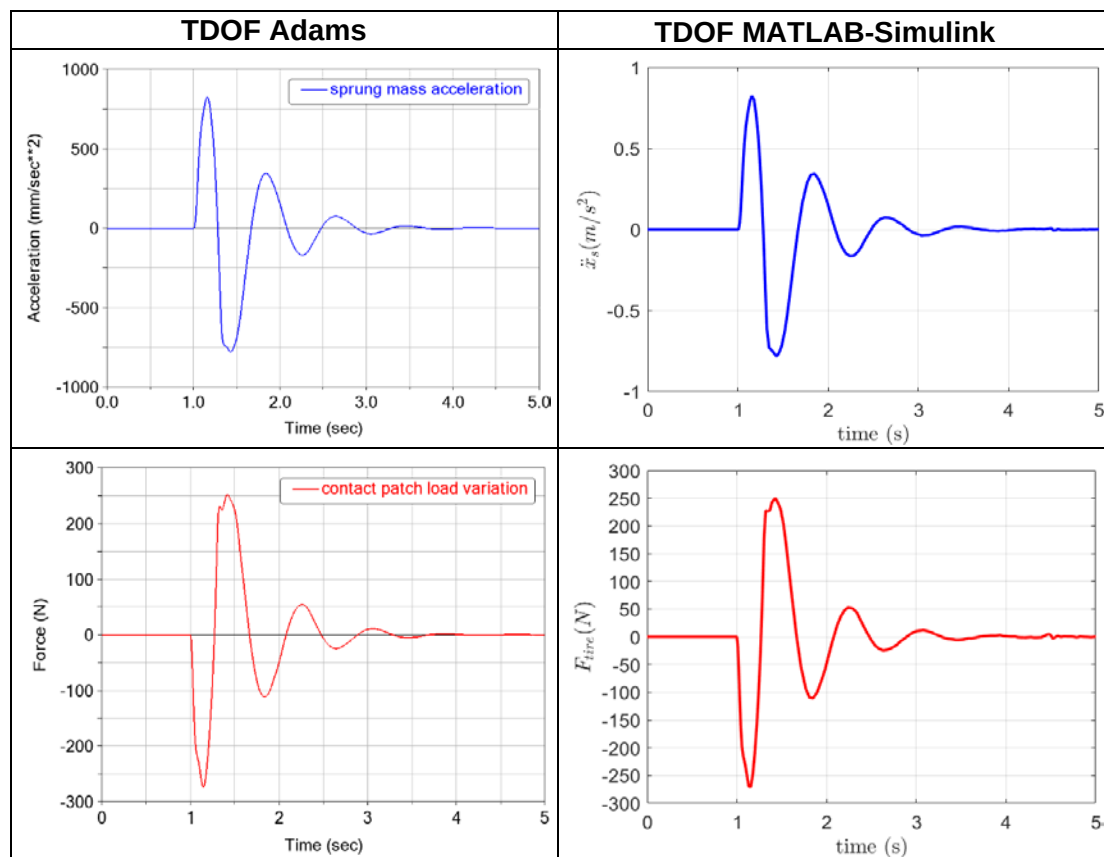
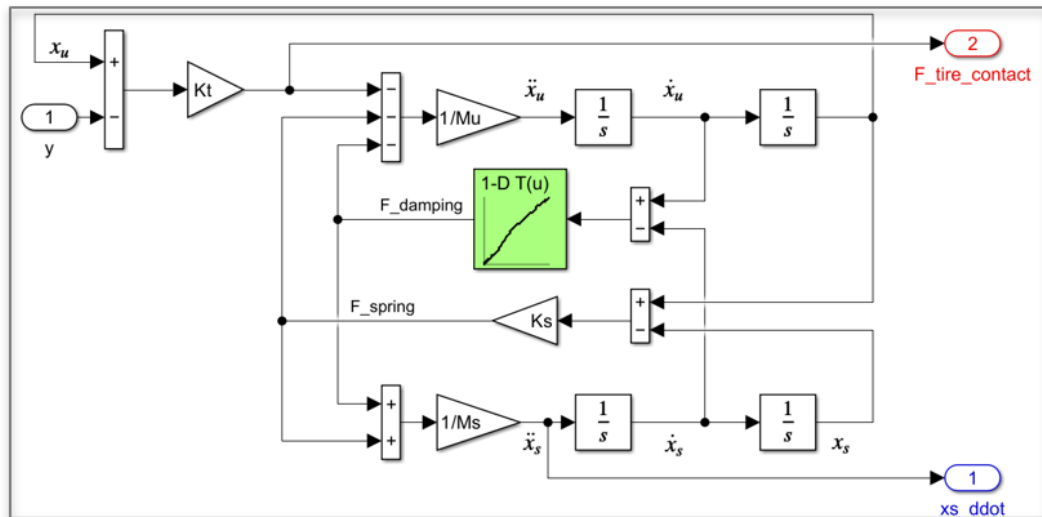


Figure 11: TDOF model frequency analysis

To increase the complexity of the model, a non-linear damping curve (obtained from the testing session on the damper dynamometer) is also incorporated into the ADAMS model through the Force-Definition method: the damping force is introduced as an external force applied between sprung and unsprung masses. The non-linear 'Low damping' curve can also be extended to the Simulink model through a 1-D LUT, implementing Eq.(20) and Eq.(21) and providing the usual step input.





Having the model accuracy verified across two different simulation environments (as detailed in Appendix D), a multivariable optimization of the load variation can be performed, acting on spring stiffness k and scaling factor c_{factor} .

on MATLAB, finding the first natural frequency after that free vibration condition is imposed (code in Appendix E).

- The scaling factor c_{factor} is introduced to account for changing the slope of the damping curve, as it can be visualized from Fig.13.

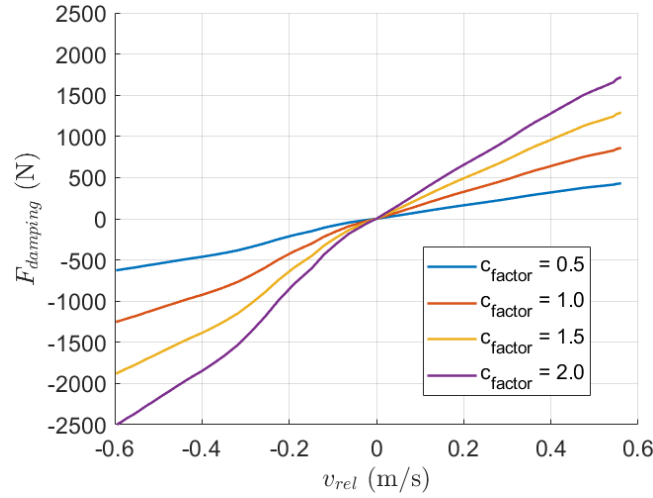
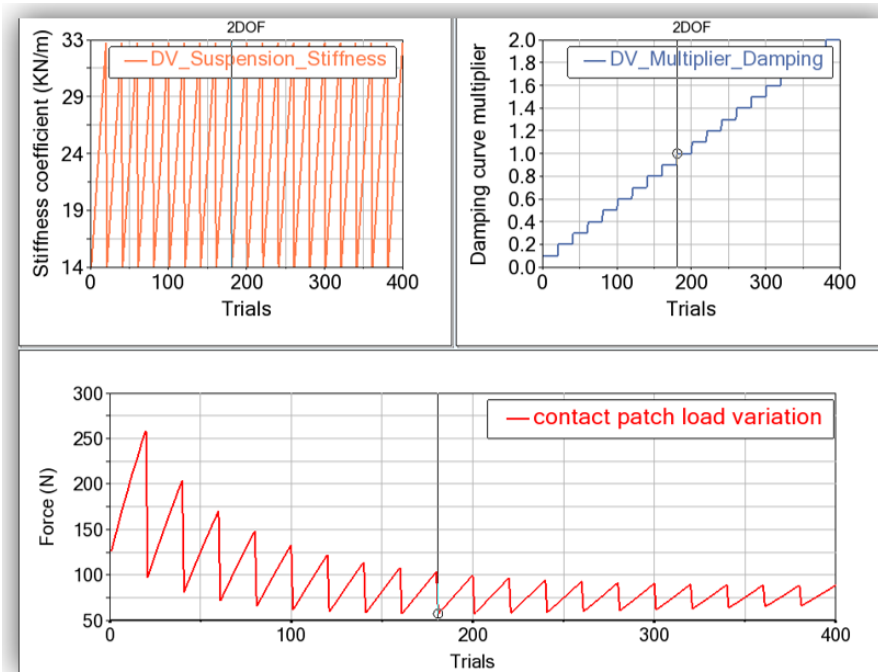


Figure 13: Scaling factor effect on 'Low damping' curve

These assumptions lead to the following boundary values for the Design of Experiments procedure:

design variable	from	up to	default levels
k_s	14 KN/m	33 KN/m	20
c_{factor}	0.1	2	20

Table 12: DOE parameters



k_s (N/m)	c_{factor} (-)	$F_{tire,RMS}$ (N)
14000	1	57.09

Figure 14: DOE results

The same optimization procedure performed in ADAMS can also be implemented in MATLAB (code in Appendix E). This approach facilitates the

optimization of the vertical acceleration of the sprung mass as well, visualizing the effect of k_s and c_{factor} variability through three-dimensional maps, from which the minimum RMS values of both force and acceleration can be extracted.

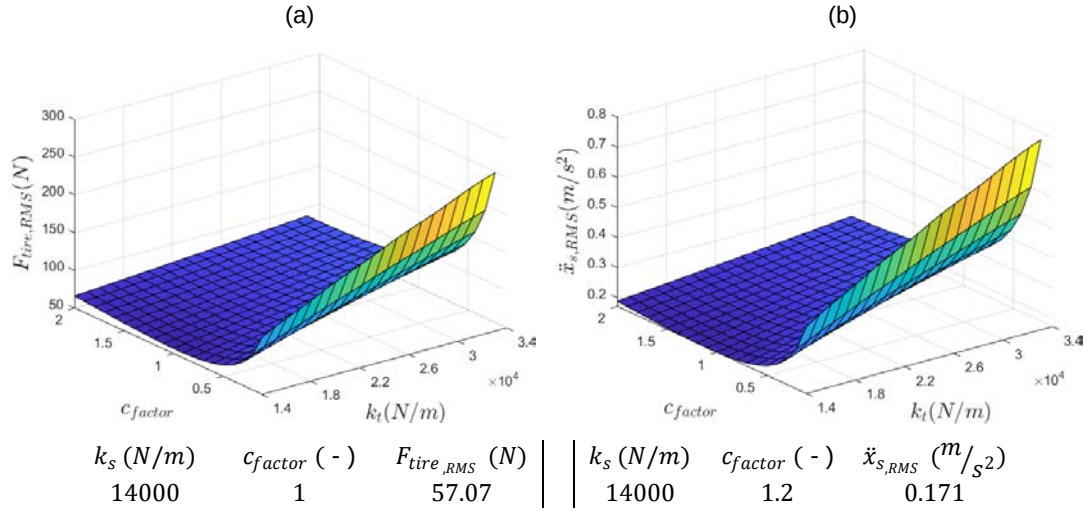


Figure 15: Load variation at contact patch (a) and vertical acceleration (b) optimization maps

It can be observed from Fig.15 that the optimization of these two quantities is achieved for two different values of c_{factor} and the lowest value of k_s , high enough to guarantee a first natural frequency higher than 1Hz.

For this reason, an intermediate value of $c_{factor} = 1.1$ may represent the most appropriate choice to ensure a suitable trade-off between performance and comfort.

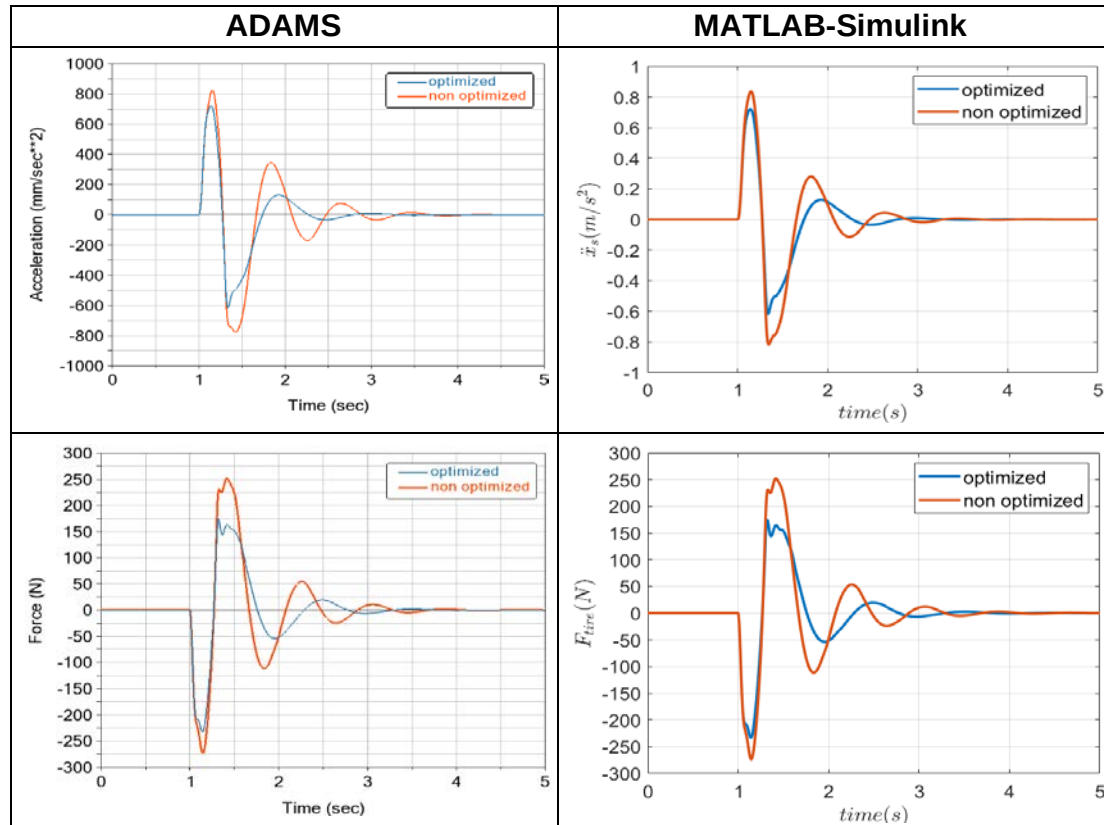


Table 13: Load at contact patch and vertical acceleration optimization

Improvements related to the adoption of optimized parameters can be visualized in Table 13, comparing force and acceleration responses to the non-optimized profiles from Table 10.

It can be observed that the optimized values of k_s and c_{factor} provide a better attenuation of both force at tire contact and acceleration response, reducing the RMS of the signal (numerically detailed in Appendix D).

2.3 Half car model (4DOF)

For sake of completeness, a model of Ford Granada with additional DOF is also analysed: the half car model allows to separate the front axle from the rear one, accounting for pitch and vertical motion of the vehicle body.

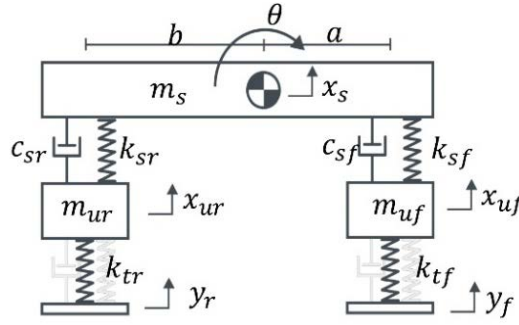


Figure 16: Half car model (Morrey, 2025)

The system behaviour can be defined through the equations of motions:

$$m_s \ddot{x}_s + c_{sf}(\dot{x}_s - \dot{x}_{uf} - a\dot{\theta}) + c_{sr}(\dot{x}_s - \dot{x}_{ur} + b\dot{\theta}) + k_{sf}(x_s - x_{uf} - a\theta) + k_{sr}(x_s - x_{ur} + b\theta) = 0 \quad (22)$$

$$I_y \ddot{\theta} - a[c_{sf}(\dot{x}_s - \dot{x}_{uf} - a\dot{\theta}) + k_{sf}(x_s - x_{uf} - a\theta)] + b[c_{sr}(\dot{x}_s - \dot{x}_{ur} + b\dot{\theta}) + k_{sr}(x_s - x_{ur} + b\theta)] = 0 \quad (23)$$

$$m_{uf} \ddot{x}_{uf} - c_{sf}(\dot{x}_s - \dot{x}_{uf} - a\dot{\theta}) - k_{sf}(x_s - x_{uf} - a\theta) + k_{tf}(x_{uf} - y_f) = 0 \quad (24)$$

$$m_{ur} \ddot{x}_{ur} - c_{sr}(\dot{x}_s - \dot{x}_{ur} + b\dot{\theta}) - k_{sr}(x_s - x_{ur} + b\theta) + k_{tr}(x_{ur} - y_r) = 0 \quad (25)$$

As previously done for the TDOF model, this set of equations can be organized in state space form on MATLAB, using the command *bode()* to show how the system behaves at different frequencies.

The analysis was carried out on heave and pitch accelerations by varying the damping coefficient, as shown in Fig.17. The peaks in the magnitude plot correspond to the natural frequencies of the system, which can be identified imposing free vibration conditions (damping equal to zero).

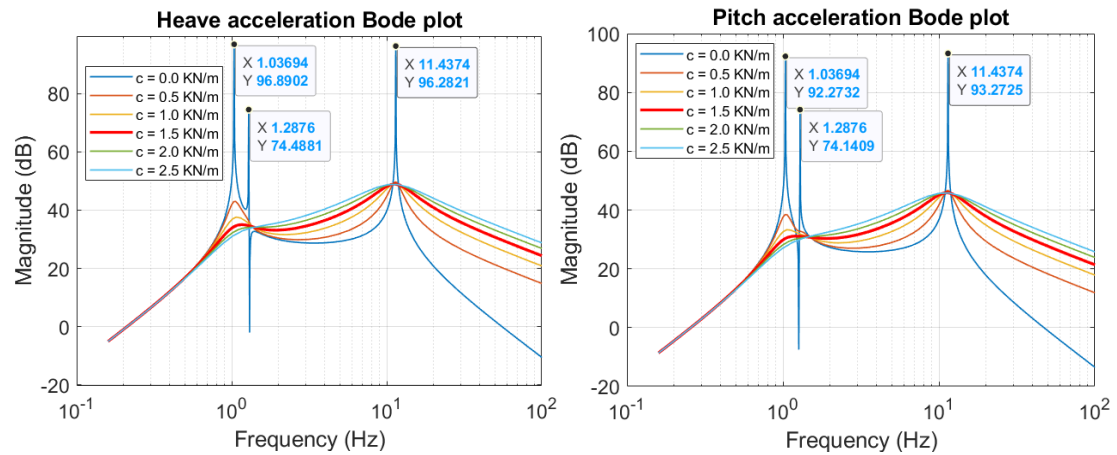


Figure 17: Half-car model frequency analysis

As further extension, the half-car model was implemented with the aim of verifying whether the parameters previously optimized for the TDOF model (applied here both at front and rear axles) yield a 4DOF model whose response to a step input is more effectively attenuated. For this purpose, a Simulink implementation is considerably more time-efficient than developing the corresponding model in ADAMS. To ensure that the bump is properly modelled, the step input on the rear axle is set equal to the one at front axle, with an additional time delay determined by the ratio between the wheelbase and the travelling speed (supposed to be 15m/s).

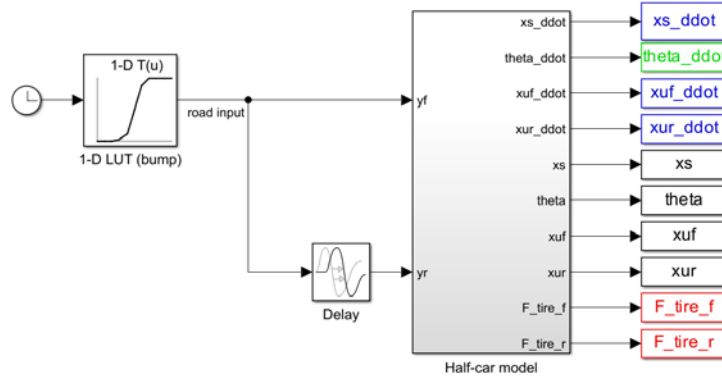


Figure 18: Half car model implementation

The ‘Low damping’ curve was additionally included inside the model to account for a non-linear damping.

The comparison between the original half car model (whose parameters are listed in Appendix C) and the one with optimized parameters (k_{sf} , k_{sr} and c_{factor} obtained from chapter 2.2) is shown below.

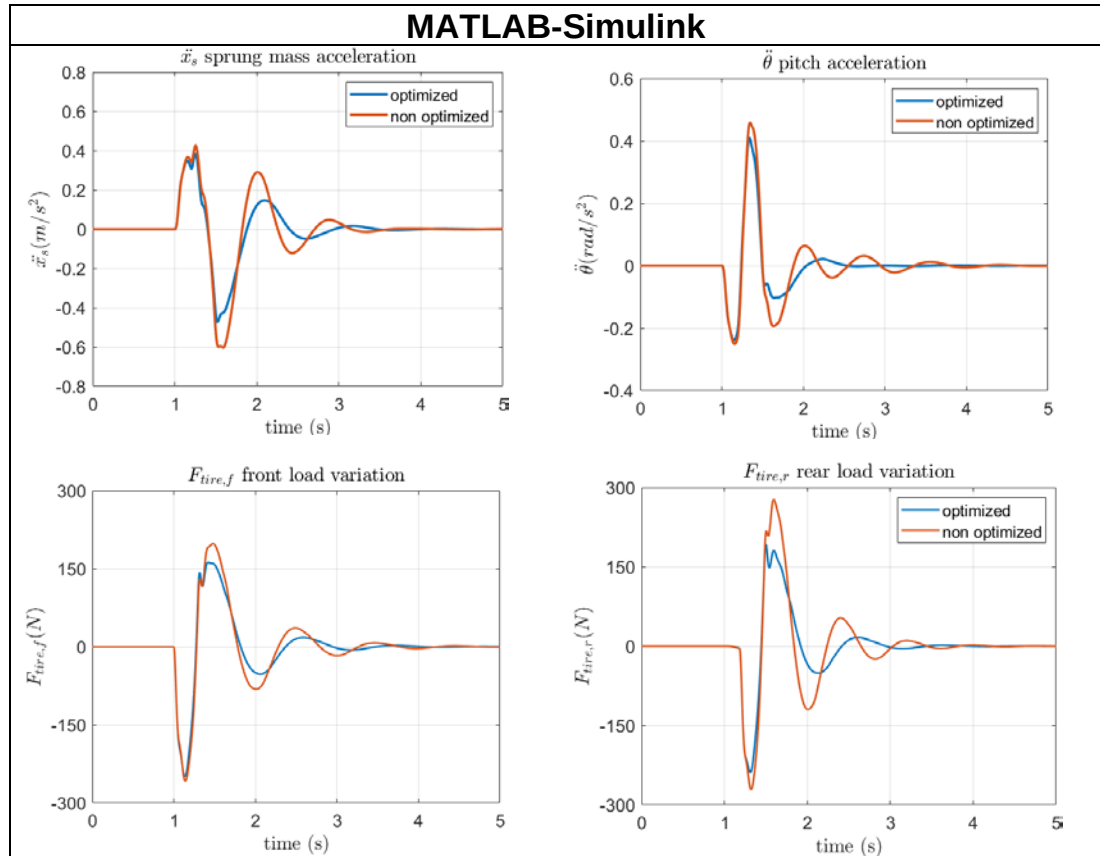


Table 14: Half car model optimization

By examining fundamental outputs such as vertical and pitching acceleration of the vehicle body and forces at tire contact for both axles, it can be observed from Table 14 that the response is more efficiently attenuated when optimized parameters from TDOF are used, with RMS values pointed out in Appendix D.

2.5 Technical conclusions

Previous analysis confirmed that progressively refined models can offer a computationally efficient framework for preliminary ride optimization.

The SDOF model proved useful for identifying general trends of vertical acceleration and load variation at tire contact while the TDOF enabled sprung-unsprung interaction, leading to the conclusion that optimization of performance rather than comfort (and vice versa) is achieved through different values for the same parameters.

This helps to prove that the aim of a vehicle dynamicist is to ensure a proper trade-off between different aspects, ultimately making decisions based on which one of them needs to be prioritized.

Finally, the half-car model demonstrated that optimised quarter-car parameters scale effectively to multi-DOF systems.

Ride performance was evaluated in terms of RMS values of load variation and body vertical acceleration, with results (listed in Appendix D) showing that up to 25% reduction can be achieved through the optimization of the models.

This procedure has proven to be efficient in the literature (Soliman, Moustafa and Shogae, 2008) with results accuracy supported by the fact that the RMS values obtained for the vertical accelerations are comparable to those reported in Fig.19 and are far from the “uncomfortable reactions” threshold according to ISO 8608. This corresponds exactly to the expected response to a 25mm bump when real vehicle parameters are used.

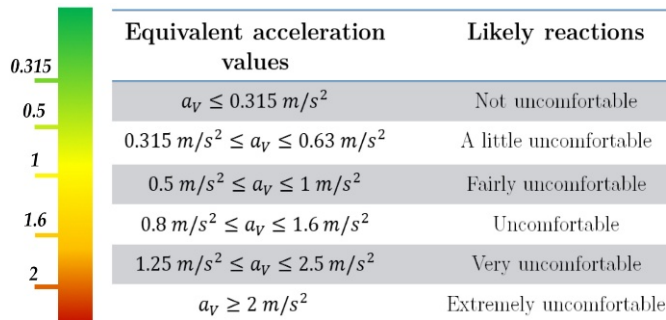


Figure 19: Levels of discomfort according to ISO 8608 (Amati, 2025)

Nevertheless, even the previously discussed models are unable to capture cross-axle coupling and full-vehicle modal interactions, requiring a model with a higher number of DOFs to account for these aspects.

(Words count: 3298)

Individual reflections

This coursework has provided a rigorous framework for deepening my understanding of vehicle vertical and lateral dynamics.

The need to validate every result across two platforms encouraged a methodological rigour that goes far beyond the numerical output itself: moving continuously between ADAMS and Simulink forced me to question the assumptions behind each model, identify their limits, and understand how modelling choices affect the predictive quality of the results.

This work also exposed several improvement opportunities that could be explored in future developments.

In the handling analysis, the fidelity of the model in reproducing the dynamic behaviour of the Nissan GT-R could be enhanced by employing vehicle-specific cornering stiffness curves instead of relying on rescaled data derived from a reference study.

For the ride optimisation, a natural extension would be to abandon the idealized road-step excitation and replace it with realistic road profiles, accounting for the vertical vibration exposure filter (ISO 2631) through its transfer function: this would allow optimisation for actual human-perceived comfort under representative manoeuvres.

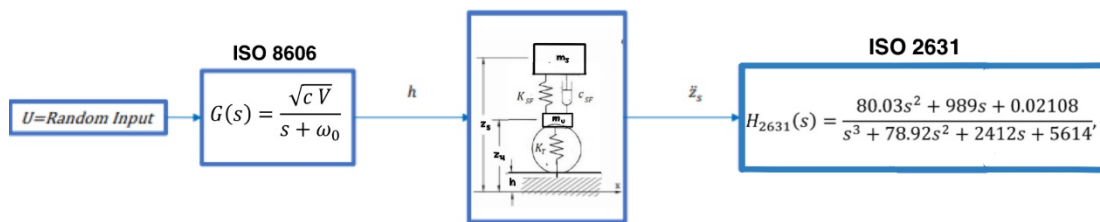


Figure 20: procedure for comfort level evaluation (Amati, 2025)

Overall, this project strengthened the awareness that simulation is not an end in itself: it is a tool whose value emerges only when the engineer is willing to question, compare, refine and iterate.

(Words count: 199)

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Appendix A: input data for bicycle model and handling analysis

Input parameters for Nissan GT-R Nismo **linear** bicycle model



	US (original model)	
Curb weight (m)	1739	kg
Overall length (L)	4690	mm
Wheelbase (l)	2780	mm
Overall height (h)	1370	mm
Overall width (W)	1895	mm
Track width (t)	1600	mm
Drag coefficient	0.26	-
Original load distribution	54% front - 46% rear	-

Table I : Nissan GT-R Nismo dataset and assumed values (Nissan Motor Co., 2017)

Yaw Inertia	Noon-1 method (MacInnis, 1997) $I_{zz} = \frac{mL^2}{12} = 3.19\text{E}+09 \text{ kg mm}^2$
C_f, C_r	Cornering stiffness model from previous coursework was normalized on MATLAB using front and rear cornering power listed in Table III as reference values, specified from a Nissan technical paper, for the model to be as realistic as possible.

Table II: Assumptions

Parameter	Symbol	unit	Value
Vehicle Mass	m	kg	1670
Vehicle Yaw Inertia	I	kgm ²	2600
Front axle ~ C.G.	l _f	m	1.08
Rear axle ~ C.G.	l _r	m	1.62
Front Cornering Power	K _f	kN/rad	62.5
Rear Cornering Power	K _r	kN/rad	95.5
Steering Gear Ratio	-	-	15.0

Table III: Parameters of a reference Nissan vehicle (Nissan Technical Review No.87, 2021)

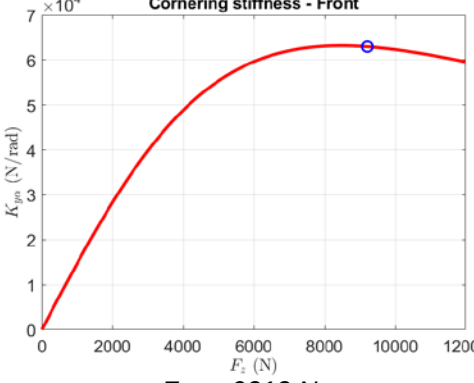
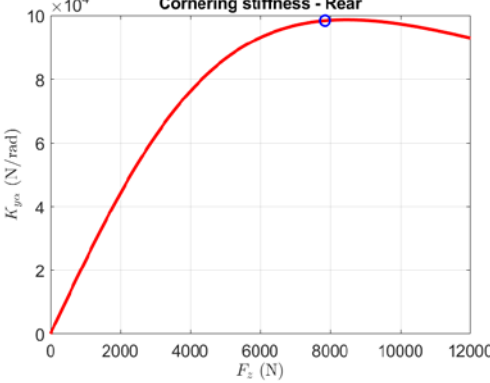
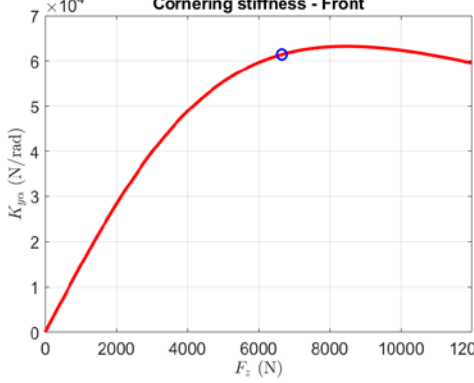
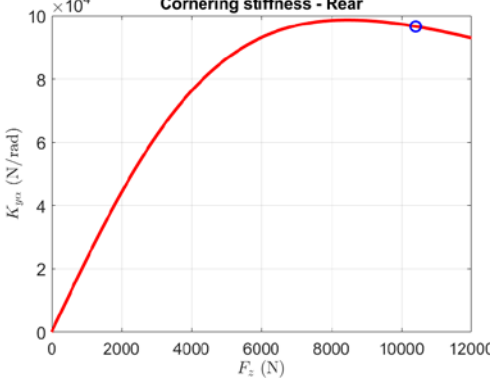
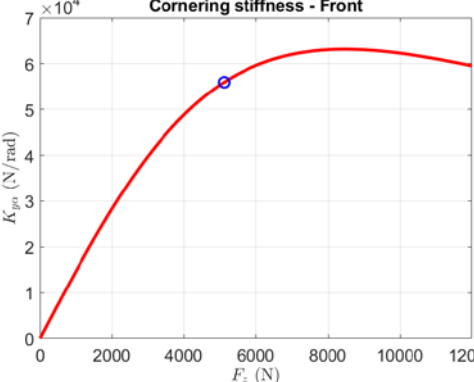
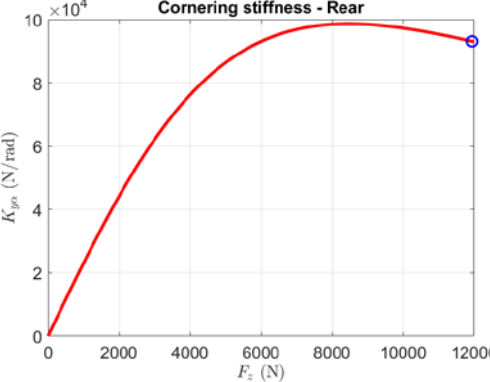
US	 Cornering stiffness - Front K_{gs} (N/rad) $\times 10^4$ F_z (N) $F_{z_f} = 9212 \text{ N}$ $C_f = 62973 \text{ N/rad (1099 N/deg)}$	 Cornering stiffness - Rear K_{gs} (N/rad) $\times 10^4$ F_z (N) $F_{z_r} = 7847 \text{ N}$ $C_r = 98369 \text{ N/rad (1717 N/deg)}$
NS	 Cornering stiffness - Front K_{gs} (N/rad) $\times 10^4$ F_z (N) $F_{z_f} = 5553 \text{ N}$ $C_f = 61404 \text{ N/rad (1072 N/deg)}$	 Cornering stiffness - Rear K_{gs} (N/rad) $\times 10^4$ F_z (N) $F_{z_r} = 10406 \text{ N}$ $C_r = 96592 \text{ N/rad (1686 N/deg)}$
OS	 Cornering stiffness - Front K_{gs} (N/rad) $\times 10^4$ F_z (N) $F_{z_f} = 5118 \text{ N}$ $C_f = 55954 \text{ N/rad (977 N/deg)}$	 Cornering stiffness - Rear K_{gs} (N/rad) $\times 10^4$ F_z (N) $F_{z_r} = 11942 \text{ N}$ $C_r = 93096 \text{ N/rad (1625 N/deg)}$

Table IV: cornering stiffness model obtained from tyre modelling coursework and rescaled according to cornering parameters of a Nissan vehicle highlighted in Table III, with code shown in Appendix E.

Appendix B: results for bicycle model and handling analysis

Linear tyre model results values for Nissan GT-R on Adams

U =15 m/s , $\delta = -15\text{deg}$	US	NS	OS	
r	-50.60	-76.57	-99.88	deg/s
α_f	-11.45	-12.40	-13.24	deg
α_r	-6.03	-12.14	-17.90	deg
$F_{y,f}$	12581	13230	12930	N
$F_{y,r}$	10354	20401	29089	N

1.2 Linear comparative analysis on MATLAB-Simulink for steady state results

U =15 m/s , $\delta = -15\text{deg}$	US linear (Adams)	US linear (MATLAB-Simulink)	
r	-50.60	-52.25	deg/s
α_f	-11.45	-11.69	deg
α_r	-6.03	-6.37	deg
$F_{y,f}$	12581	12847	N
$F_{y,r}$	10354	10944	N

1.4 Non-linear tyre model results comparison for Nissan GT-R

U =15 m/s , $\delta = -15\text{deg}$	US linear (Adams)	US non-linear (Adams)	
r	-50.60	-8.93	deg/s
α_f	-11.45	-14.43	deg
α_r	-6.03	-1.50	deg
$F_{y,f}$	12581	2267	N
$F_{y,r}$	10354	1866	N

1.5 Non-linear comparative analysis on MATLAB-Simulink for steady state results

U =15 m/s , $\delta = -15\text{deg}$	US non-linear (Adams)	US non-linear (MATLAB-Simulink)	
r	-8.93	-9.22	deg/s
α_f	-14.43	-14.43	deg
α_r	-1.50	-1.14	deg
$F_{y,f}$	2267	2270	N
$F_{y,r}$	1866	1934	N

1.3 Derivative analysis results

Input: $U = 15 \text{ m/s}$, $\delta = -15 \frac{\pi}{180} \text{ rad}$

Derivatives			Derived parameters (underdamped)		
Y_β	-161342	N / rad	k	189496	
Y_r	4476	Ns / rad	c	41375	
Y_δ	62973	N / rad	ω_n	7.71 (1.23)	rad/s (Hz)
N_β	67146	Nm / rad	c_{crit}	49173	
N_r	-21644	$Nm \cdot s / \text{rad}$	ζ	0.841	-
N_δ	80530	Nm / rad	ω_d	4.16 (0.66)	rad/s (Hz)
			$C2$	660186	
			Φ	0.4954	
			X	-1.9185	
			r_{ss}	52.25	deg/s

Appendix C: input parameters for suspension dynamics and ride optimisation



Input parameters of Ford Granada 2DOF model

Body mass (m_s)	317.5	kg
Wheel mass (m_u)	45.4	kg
Tire stiffness coefficient (k_t)	192	kN/m
Spring stiffness coefficient (k_s)	22	kN/m
Damping coefficient (c_s)	1.5	kNs/m

Table V: Ford Granada 2DOF dataset (Crolla, 1991)

Input parameters of 1DOF model are derived from 2DOF model:

Vehicle mass (m)	assumed to be equal to the sprung mass of the quarter car 2DOF $m = 317.5$	kg
Suspension stiffness (k)	assumed as equivalent stiffness of two springs in series from the 2DOF: $k = \left(\frac{1}{k_s} + \frac{1}{k_t} \right)^{-1} = 19.74$	kN/m
Damping coefficient (c)	1.5	kNs/m

Table VI: derived Ford Granada 1DOF lumped parameters

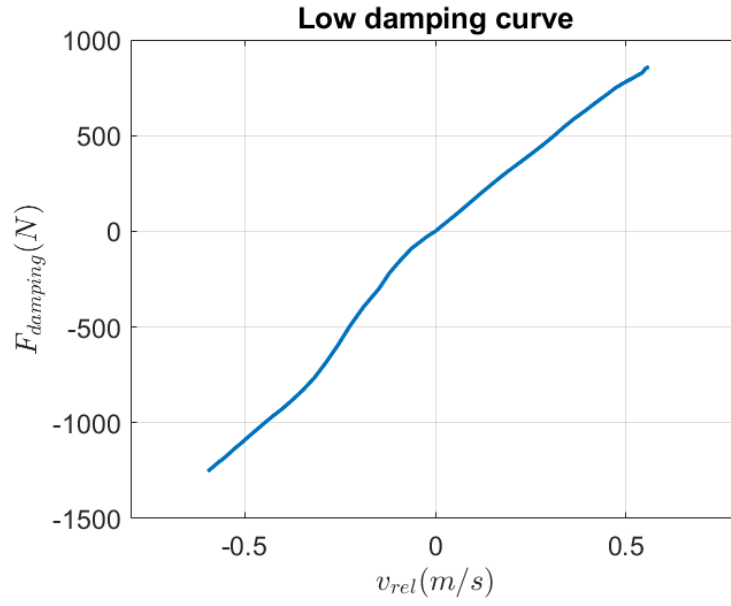


Figure I: Damper dynamometer 'Low damping' curve implemented inside the models

Input parameters of Ford Granada Half car model

<u>Masses and Inertias</u>		
M_{bh}	0.690	Half vehicle body mass (tonnes)
I_b	1.222	Pitch inertia (tonnes.m ²)
M_{wf}	0.0405	Front wheel mass (tonnes)
M_{wr}	0.0454	Rear wheel mass (tonnes)
<u>Stiffness and Damping</u> (single wheel station values)		
K_{tf}	192	Front tyre stiffness (kN/m)
K_{tr}	192	Rear tyre stiffness (kN/m)
K_{sf}	17	Front susp. stiffness (kN/m)
K_{sr}	22	Rear susp. stiffness (kN/m)
C_{sf}	1.5	Front damping coeff. (kNs/m)
C_{sr}	1.5	Rear damping coeff. (kNs/m)
<u>Geometry</u>		
a	1.25	Dist. from cg to front wheel station (m)
b	1.51	Dist. from cg to rear wheel station (m)
l	2.76	Wheelbase (m)

Table VII: Ford Granada Half car dataset (Crolla, 1991)

Appendix D: results for suspension dynamics and ride optimisation

SDOF model results for Ford Granada (linear damping)

Non optimized: $k_s = 22 \text{ KN/m}$, $c = 1.5 \text{ kN} \cdot \text{s/m}$

$y_{ss} = 25 \text{ mm}$	1DOF (Adams)	1DOF (MATLAB/Simulink)	
$\ddot{x}_{RMS}$	0.209	0.207	$\frac{m}{s^2}$
$F_{tire,RMS}$	64.64	64.49	N

Error! Reference source not found. results for Ford Granada (non-linear damping)

Non optimized: $k_s = 22 \text{ KN/m}$, $c_{factor} = 1$

Optimized: $k_s = 14 \text{ KN/m}$, $c_{factor} = 1.1$

$y_{ss}=25 \text{ mm}$	2DOF (Adams)		2DOF (MATLAB/Simulink)		
	Non optimized	Optimized	Non optimized	Optimized	
$\ddot{x}_{s,RMS}$	0.230	0.171 (-25.6%)	0.227	0.171 (-24.7%)	$\frac{m}{s^2}$
$F_{tire,RMS}$	74.90	57.39 (-23.4%)	74.84	57.37 (-23.4%)	N

Half car model results for Ford Granada (non-linear damping)

Non optimized: $k_{s,f} = 17 \text{ KN/m}$, $k_{s,r} = 22 \text{ KN/m}$, $c_{factor} = 1$

Optimized: $k_{s,f} = k_{s,r} = 14 \text{ KN/m}$, $c_{factor} = 1.1$

$y_{ss} = 25 \text{ mm}$	4DOF (MATLAB/Simulink)		
	Non optimized	Optimized	
$\ddot{x}_{s,RMS}$	0.162	0.125 (-22.8%)	$\frac{m}{s^2}$
$\ddot{\theta}_{RMS}$	0.0943	0.0793 (-15.9%)	$\frac{rad}{s^2}$
$F_{tire,RMS} (front)$	66.29	58.64 (-11.5%)	N
$F_{tire,RMS} (rear)$	75.90	57.43 (-24.4%)	N

Appendix E: MATLAB script

Cornering stiffness evaluation for linear bicycle model (US) - Table IV in Appendix A

```
m = 1739; % Nissan GTR vehicle mass (kg)
m_data = 1670; %from Nissan table: reference Vehicle mass (kg)
a1 = 1.08; %from Nissan table: front axle to CoG (m)
b1 = 1.62; %from Nissan table: rear axle to CoG (m)
g = 9.81; % gravity acceleration (m/s^2)

% Cornering stiffness curve from previous coursework rescaled using Nissan
paper values
load('Kya_Fz.mat');
Fz_fit = Fz_fit';
Kya_fit = Kya_fit';

% Front scaling
KyaF_data = 62500; %from Nissan table: Front Cornering power (N/rad)
FzF_data = m_data * g * b1/(a1+b1); %from Nissan table: Front Load (N)
FzF_GTR = (1-0.46) * m * g; %Front Load Nissan GTR (N)
KyaF_norm = normalize(Kya_fit, 'range');
KyaF_data_norm = interp1(Fz_fit, KyaF_norm, FzF_data, 'linear');
KyaF_scaled = KyaF_norm * KyaF_data/KyaF_data_norm;
Cf = interp1(Fz_fit, KyaF_scaled, FzF_GTR, 'linear'); % (N/rad)

% Plot
figure
plot(Fz_fit, KyaF_scaled, 'r', LineWidth=3);
hold on
plot(FzF_GTR, Cf, 'bo', 'MarkerSize', 10, 'LineWidth', 2);
ylabel('$K_{y \alpha}$ (N/rad)', Interpreter='latex');
xlabel('$F_z$ (N)', Interpreter='latex');
title('Cornering stiffness - Front');
grid on;

% Rear scaling
KyaR_data = 95500; %from Nissan table: Front Cornering power (N/rad)
FzR_data = m_data * g * a/(a+b); %from Nissan table: Rear Load (N)
FzR_GTR = 0.46 * m * g; %Rear Load Nissan GTR (N)
KyaR_norm = normalize(Kya_fit, 'range');
KyaR_data_norm = interp1(Fz_fit, KyaR_norm, FzR_data, 'linear');
KyaR_scaled = KyaR_norm * KyaR_data/KyaR_data_norm;
Cr = interp1(Fz_fit, KyaR_scaled, FzR_GTR, 'linear'); % (N/rad)

% Plot
figure
plot(Fz_fit, KyaR_scaled, 'r', LineWidth=3);
hold on
plot(FzR_GTR, Cr, 'bo', 'MarkerSize', 10, 'LineWidth', 2);
xlabel('$F_z$ (N)', Interpreter='latex');
ylabel('$K_{y \alpha}$ (N/rad)', Interpreter='latex');
title('Cornering stiffness - Rear');
grid on;
```

State space system for linear bicycle model (US) - Fig.2, Fig.4, Fig.5 and Table 3

% Vehicle parameters below need to be changed to study also the NS and OS configurations

% Nissan GT-R parameters

```
m = 1739; % Nissan GTR vehicle mass (kg)
l = 2.780; % wheelbase (m)
a = 0.46*l; % front axle to CG (m)
b = (1-0.46)*l; % rear axle to CG (m)
U = 15; % tested longitudinal speed (m/s)
I = 3190; % yaw inertia (kg*m^2)
```

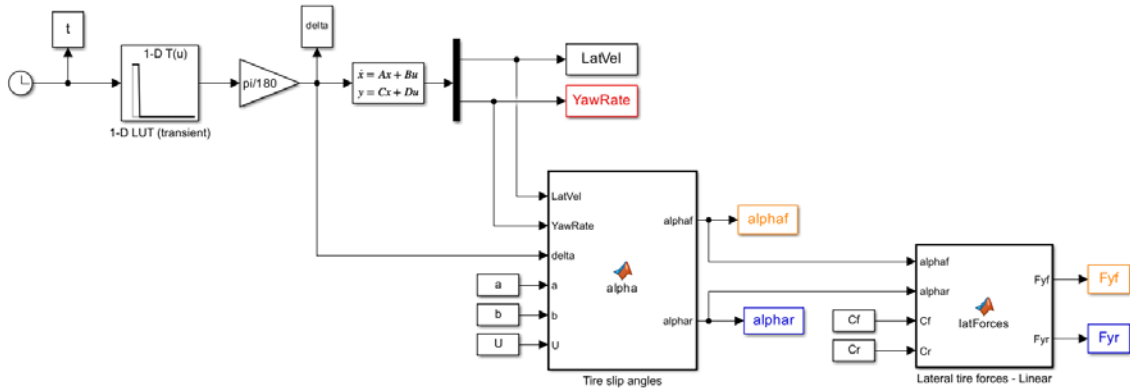
% Bicycle model state space matrices

```
A = [-(Cf + Cr)/(m*U) -(U + (a*Cf-b*Cr)/(m*U));
      -(a*Cf-b*Cr)/(I*U) -(a^2*Cf+b^2*Cr)/(I*U)];
B = [Cf/m; a*Cf/I];
C = eye(2)
D = [0;0]
```

%Sim

```
Tsim = 10;
t = 0:0.1:Tsim; % Step Steer (accounting for transient)
delta = interp1([ 0, 1, 1.1, Tsim],[0, 0, -15, -15],t);
```

```
file = "Linear_Bicycle_model.slx";
sim(file);
```



% Understeer gradient

```
Kus = m/l * (b/Cf - a/Cr);
```

%% Plots

%Yaw Rate

```
figure;
plot(YawRate.Time, YawRate.data *180/pi, LineWidth=2, Color='r');
xlabel('time (s)');
ylabel('Yaw Rate (deg/s)');
legend('Yaw Rate');
grid on;
```

% Tire slip angles

```
figure;
```

```

plot(alphar.time, alphar.data, LineWidth=2);
hold on;
plot(alphaf.time, alphaf.data, LineWidth=2);
xlabel('time (s)');
ylabel('Tire Slip Angle (deg)');
legend('\alpha_r', '\alpha_f');
grid on;

% Lateral forces
figure;
plot(Fyr.time, Fyr.data, LineWidth=2);
hold on;
plot(Fyf.time, Fyf.data, LineWidth=2);
xlabel('time (s)');
ylabel('Lateral Force (N)');
legend('F_{yr}', 'F_{yf}');
grid on;

% Curvature gain - Fig.2
figure;
for Kus = [0.0068, 6.3e-05, -0.0038]
    V = linspace(0, 50, 5000);
    curvGain = 1./(1 + Kus * V.^2);
    plot(V, curvGain, LineWidth=2);
    hold on
    ylim([-5,5])
end
grid on;
xlabel('Vehicle Speed (m/s)', FontSize=28);
ylabel('$\frac{1}{R} \delta_f$', 'Interpreter','latex', FontSize=28);
legend('US', 'NS', 'OS', FontSize=28)

```

Spline development for Nissan GT-R (US) - Table 4

```

% Nonlinear Tire model
splineData = readmatrix("Spline.xlsx");
alpha = splineData(:,1);
Fy_norm = splineData(:,2);
alphaSpace = linspace(-90,90,10000);
s = spline(alpha, Fy_norm, alphaSpace);
FyR = s * (Cr * pi/180); % Cr in (N/deg)
FyF = s * (Cf * pi/180); % Cf in (N/deg)

figure;
plot(alphaSpace, FyF, 'Color', '#FFA500', LineWidth=2 );
grid on
xlabel('Tire slip angle (deg)');
ylabel('$F_{yf}$ (N)', Interpreter='latex');
title('Non-linear Front cornering model');

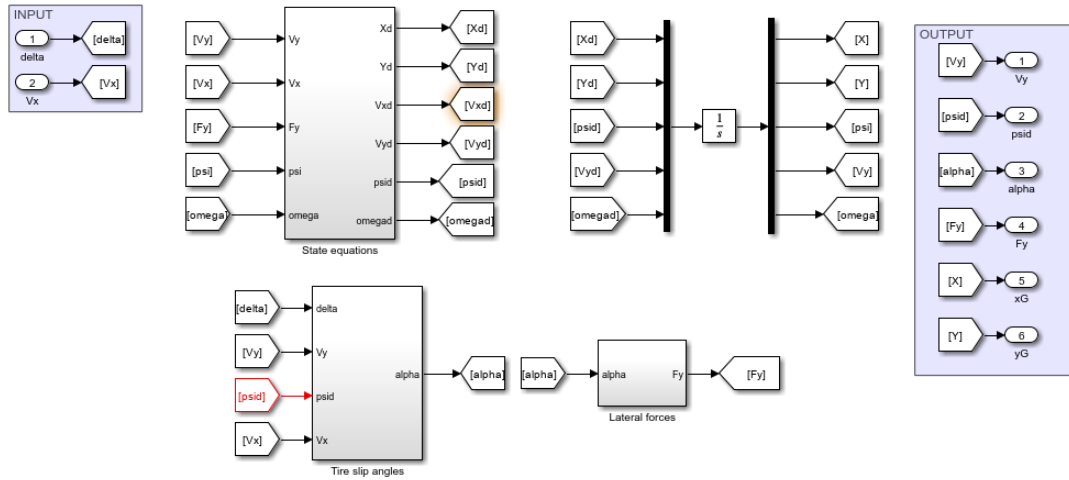
figure;
plot(alphaSpace, FyR, 'b', LineWidth=2);
grid on
xlabel('Tire slip angle (deg)');
ylabel('$F_{yr}$ (N)', Interpreter='latex');
title('Non-linear Rear cornering model');

```

Non-linear bicycle model (US) - Fig.8 and Table 6

```
FyF_spline = FyF';
FyR_spline = FyR';
alpha_spline = alphaSpace';
% Same simulation settings used for linear model
% Same step steering input
zita0 = [0, 0, 0, 0, 0]; % Initial conditions
```

```
file = 'Nonlinear_Bicycle_model.slx';
sim(file);
```



```
figure;
plot(-yG, xG, LineWidth=2); %from local to global reference frame
xlabel('X_G (m)');
ylabel('Y_G (m)');
grid on;
```

```
figure;
plot(t, FyR, LineWidth=2);
hold on;
plot(t, FyF, LineWidth=2);
xlabel('time (s)');
ylabel('Lateral Force (N)');
legend('F_{yr}', 'F_{yf}');
grid on;
```

```
figure;
plot(t, alphaR, LineWidth=2);
hold on;
plot(t, alphaF, LineWidth=2);
xlabel('time (s)');
ylabel('Side-Slip Angle (deg)');
legend('\alpha_r', '\alpha_f');
grid on;
```

```
figure
plot(t, psid*180/pi, 'LineWidth', 2, 'Color', 'r');
ylabel('Yaw Rate (deg/s)');
ylim([-12,0])
xlabel('Time (s)');
legend('Yaw Rate');
```

Derivative analysis (US) - Fig.7

```
%input parameters US configuration

Cf = 62973;           % front cornering stiffness US (N/rad)
Cr = 98369;           % rear cornering stiffness US (N/rad)
l = 2.780;            % wheelbase (m)
a = 0.46*l;           % CG - front axle (m)
b = 0.54*l;           % CG - rear axle (m)
m = 1739;             % vehicle mass (kg)
I = 3190;             % yaw inertia (kgm^2)

%input parameters for underdamped response
U = 15;               % vehicle speed (m/s)
delta = 15*pi/180;    % steering input (rad)

% Derivatives
Y_beta = -Cf - Cr;    %Damping in side slip
Y_r = -(a*Cf - b*Cr)./U; %Lateral force/ yaw coupling
Y_deltaf = Cf;        %Control force
N_beta = -(a*Cf - b*Cr); %Static directional stability
N_r = -(a^2 * Cf + b^2 *Cr)./U; %Yaw damping
N_deltaf = a*Cf;      %Control moment

%Derived paramters
C2 = (Y_deltaf*N_beta)/(m*U) - (N_deltaf*Y_beta)/(m*U);
c = -(N_r + I*Y_beta)/(m*U);
k = N_beta + (Y_beta*N_r)/(m*U) - (N_beta*Y_r)/(m*U);
wn = sqrt(k/I);       %Natural frequency (rad/s)
Cc = (2*I*wn);        %Critical damping
zita = c/ Cc;         %Damping ratio (-)
wd = wn*sqrt(1-zita^2); %Damped frequency (rad/s)
phi = atan(wd/wn);    %Phase difference (rad)
X = -C2*delta/(k*sin(phi)); %Amplitude

t = 0:0.01:9;
r = X*exp(-zita.*wn.*t).*sin(wd.*t + phi) + C2/k*delta;

%Yaw Rate response
figure;
plot(t, r*180/pi, LineWidth=2, Color='r');
xlabel('time (s)');
ylabel('yaw rate (deg/s)');
title('Underdamped response')
ylim([0, 60])
legend('r');
grid on;
```

SDOF vertical model simulation - Fig.10 and Table 8

```
% Ford Granada vehicle parameters
M = 317.5;           % vehicle mass (kg)
K = 19.74e3;         % spring stiffness (N/m)
c = 1.5e3;           % damping coefficient (Ns/m)

%Bump input (accounting for 0.3 seconds transient)
bump = readmatrix('step', 'NumHeaderLines', 2); % Adams road step
t = bump(3:end,2);   %s
```



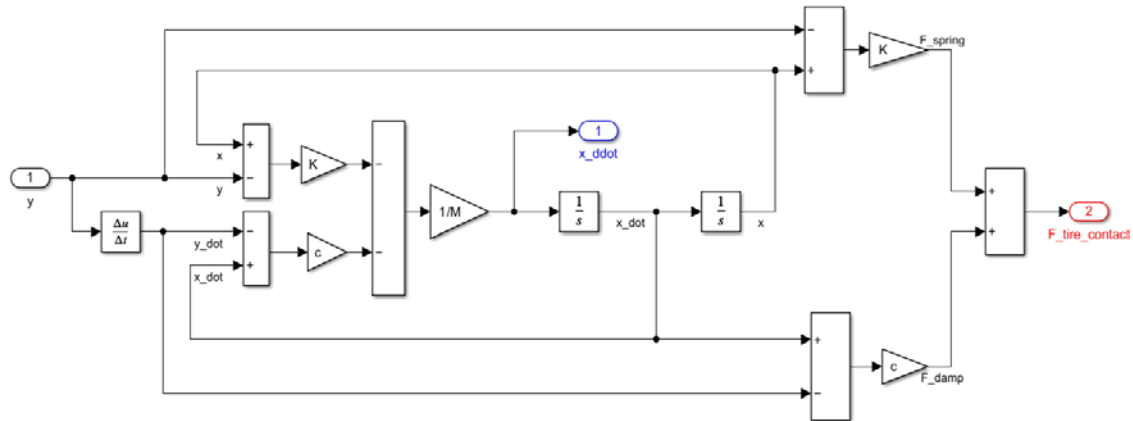
```

y = bump(3:end,1)./1000; %m

% Simulation
T_sim = 5;

file = 'Vertical_1DOF_sim.slx'; %Fig.11
sim(file);

```



```

% Plots
figure;
plot(x_ddot.time, x_ddot.data, 'b', LineWidth=2);
xlabel('Time (s)');
ylabel('acc_{y} (m/s^2)');
ylim([-1,1])
grid on;
rms(x_ddot.data)

figure
plot(F_tire.time, F_tire.data, 'r', LineWidth=2);
xlabel('Time (s)');
ylabel('F_{tire} (N)');
yticks(-250:50:250)
ylim([-250,250])
grid on;
rms(F_tire.data)

```

TDof Frequency analysis (Bode diagrams) starting from state space - Fig.11

```

clear all
close all
clc

%% Vehicle Parameters
ms = 317.5; %kg
mu = 45.4; %kg
kt = 192e3; %N/m
ks = 22e3; %N/m
c_sys = 1.5e3; %Ns/m

%% state space matrices

%x = [xu, xs, xu_dot, xs_dot]' states
%u = road input

```

```

legend_entries = {};
for zita = [0.1, 0.2, 0.3, 0.4, 0.5, 1, 1.5, c_sys/(2*sqrt(ks*ms))]

    c = zita*2*sqrt(ks*ms);

    A= [ 0          0          1          0
          0          0          0          1
        -(ks+kt)/mu  ks/mu      -c/mu      c/mu
          ks/ms      -ks/ms      c/ms      -c/ms];

    B= [ 0          0          kt/mu      0 ]';

    C= [ks/ms      -ks/ms      c/ms      -c/ms      %sprung mass acc
          kt          0          0          0];      %force at tyre contact

    D= [0      -kt]';

    %sprung mass acceleration bode
    sys_acc_s = ss(A, B, C(1,:), D(1,:));
    [mag1,~,w1] = bode(sys_acc_s); %input = front road profile
    mag1 = squeeze(mag1);
    f1 = w1 / (2*pi);
    figure(1);
    if zita == c_sys/(2*sqrt(ks*ms))
        semilogx(f1, 20*log10(mag1), 'r', LineWidth=2); % plot in Hz
        legend_entries{end+1} = sprintf('c = %.2f kN*s/m', c_sys/1000);
    else
        semilogx(f1, 20*log10(mag1), LineWidth=1); % plot in Hz
        legend_entries{end+1} = sprintf('\zeta = %.1f', zita);
    end
    xlim([1e-1,1e2])
    xlabel('Frequency (Hz)', 'FontSize',16);
    ylabel('Magnitude (dB)', 'FontSize',16);
    set(gca, 'FontSize', 14)
    title('Sprung mass acceleration - Bode plot');
    grid on;
    hold on

    %Load variation at tire contact bode
    sys_F_tire = ss(A, B, C(2,:), D(2,:));
    [mag2,~,w2] = bode(sys_F_tire); %input = front road profile
    mag2 = squeeze(mag2);
    f2 = w2 / (2*pi);
    figure(2);
    if zita == c_sys/(2*sqrt(ks*ms))
        semilogx(f2, 20*log10(mag2), 'r', LineWidth=2); % plot in Hz
    else
        semilogx(f2, 20*log10(mag2), LineWidth=1); % plot in Hz
    end
    xlim([1e-1,1e2])
    xlabel('Frequency (Hz)', 'FontSize',16);
    ylabel('Magnitude (dB)', 'FontSize',16);
    set(gca, 'FontSize', 14)
    title('Load variation at tire contact - Bode plot');
    grid on;
    hold on
end

```

```
figure(1)
legend(legend_entries, 'Location', 'best', FontSize=10);
figure(2)
legend(legend_entries, 'Location', 'best', FontSize=10);
```

TD0F vertical model simulation - Fig.12 and Table 11

```
clear all
close all
clc

% Ford Granada vehicle parameters
Mu=45.4;      % Kg
Ms=317.5;    % Kg
Kt=192e3;    % N/m    %to be optimized
Ks=22e3;     % N/m

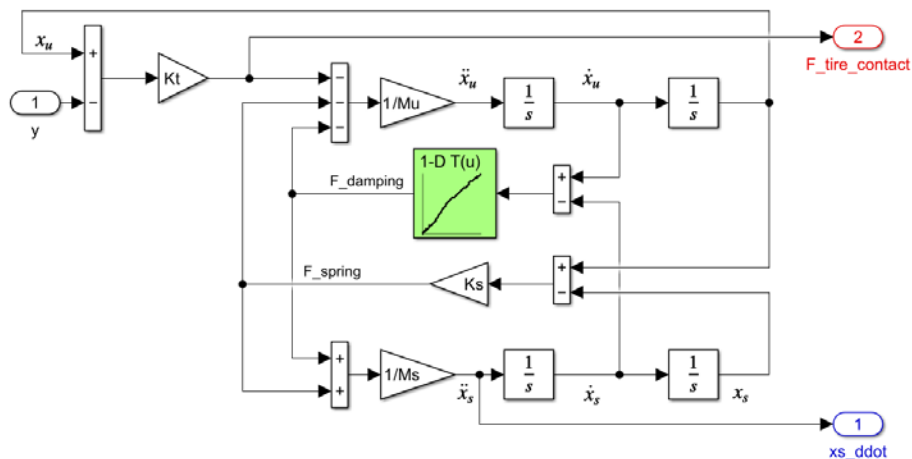
%if 1-D LUT Spline for damper-dyno curve is implemented
data = readmatrix("Damper_Low.txt");
v_relative = data(:,1)/1000;           %m/s
F_damping = data(:,2);                 %N

%% Simulation
T_sim = 5;    % s

%Bump simulation (accounting for 0.3 seconds transient)
bump = readmatrix('step', 'NumHeaderLines', 2); %step imported from Adams
t = bump(3:end,2);
y = bump(3:end,1)./1000;

file = 'Low_damping_Spline_2DOF'; %Fig.13

sim(file);
```



```
%% Figures
figure(1)
plot(xs_ddot.time,xs_ddot.data, 'b','linewidth',2)
xlabel('time [s]','fontsize',16)
ylabel('sprung mass acc_y [m/s^2]','fontsize',16)
ylim([-1,1])
grid on
```

```

rms(xs_ddot.data)

figure(2)
plot(F_tire.time,F_tire.data, 'r','linewidth',2)
xlabel('time [s]','fontsize',16)
ylabel('F_{tire} [N]','fontsize',16)
yticks(-300:50:300)
ylim([-300,300])
grid on
rms(F_tire.data)

```

TDof: evaluation of the suspension stiffness domain that ensures a first natural frequency f_1 : $1\text{Hz} < f_1 < 1.5\text{Hz}$ according to ISO 2631

```

clc
clear all
close all

% Parameters of the quarter-car model
ms = 317.5;      % sprung mass [kg]
mu = 45.4;      % unsprung mass [kg]
kt = 192000;    % tire stiffness [N/m]

for ks = 14000:1000:33000    % suspension stiffness domain [N/m]
    M = [ms    0;
          0    mu];
    K = [ ks    -ks;
          -ks    ks+kt];

    [phi, lambda] = eig(K, M); % eigenvalue problem
    lambda = diag(lambda);    % eigenvalues (w^2)
    omega = sqrt(lambda);     % Natural frequencies in rad/s
    omega = sort(omega);
    f1 = omega(1) / (2*pi);   % First natural frequency in Hz

    % results are in between 1Hz and 1.5Hz for the chosen stiffness range
    fprintf('First natural frequency f1 = %.4f Hz\n', f1);
end

```

TDof multivariable non-linear optimization - Fig.15a and Fig.15b

```

% Ford Granada vehicle parameters
Mu=45.4;      % Kg
Ms=317.5;     % Kg
Kt=192e3;     % N/m
Ks=22e3;      % N/m    %to be optimized

%1-D LUT Spline for damper-dyno curve
data = readmatrix("Damper_Low.txt");
v_relative = data(:,1)/1000;      %m/s
F_damping = data(:,2);           %N

% Bump input (accounting for 0.3s transient)
bump = readmatrix('step', 'NumHeaderLines', 2); %step imported from Adams
t = bump(3:end,2);
y = bump(3:end,1)./1000;

T_sim = 5;    % s

```

```

file = 'optimized_2DOF';

xs_ddot_rms_opt = [];
F_damp_rms_opt = [];

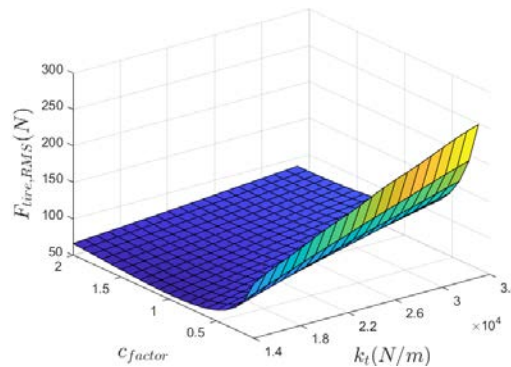
%20x20 optimization
c_multiplier = 0.1:0.1:2;           %range of optimization for c_multiplier
k = 14000:1000:33000;               %range of optimization for Ks evaluated

for i = 1:length(c_multiplier)
    F_opt = F_damping*c_multiplier(i);
    for j = 1:length(k)
        Ks = k(j);
        sim(file);
        xs_ddot_rms_opt(i,j) = rms(xs_ddot.data);
        F_damp_rms_opt(i,j) = rms(F_tire.data);
    end
    figure(1);
    hold on;
    plot(v_relative, F_opt);
end

%% 3D optimized maps plot

figure(2) %Fig.16a
surf(k, c_multiplier, F_damp_rms_opt)
xlabel('$k_t$ (N/m)', 'Interpreter', 'latex', 'FontSize', 16)
ylabel('$c_{factor}$', 'Interpreter', 'latex', 'FontSize', 16)
zlabel('$F_{tire,RMS}$ (N)', 'Interpreter', 'latex', 'FontSize', 16)
xlim([14000,34000])
xticks(14000:4000:34000)
ylim([0.1,2])

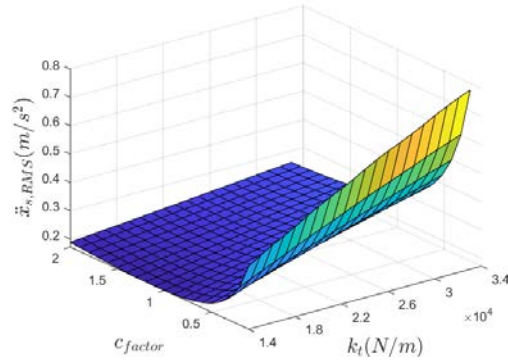
```



```

figure(3) %Fig.16b
surf(k, c_multiplier, xs_ddot_rms_opt)
xlabel('$k_t$ (N/m)', 'Interpreter', 'latex', 'FontSize', 16)
ylabel('$c_{factor}$', 'Interpreter', 'latex', 'FontSize', 16)
xlim([14000,34000])
xticks(14000:4000:34000)
ylim([0.1,2])
zlabel('$\ddot{x}_{s,RMS}$ (m/s^2)', 'Interpreter', 'latex', 'FontSize', 16)

```



TD0F Optimized and non-optimized force and acceleration comparison - Table 13

```
%% Parameters
clc
clear all
close all

Mu=45.4;      % Kg
Ms=317.5;     % Kg
Kt=192e3;     % N/m    %to be optimized

%if 1-D LUT Spline for damper-dyno curve is implemented
data = readmatrix("Damper_Low.txt");
v_relative = data(:,1)/1000;      %m/s
F_damping = data(:,2);           %N

T_sim = 5;      % s

% Bump input (accounting for transient)
bump = readmatrix('step', 'NumHeaderLines', 2); %step imported from Adams
t = bump(3:end,2);
y = bump(3:end,1)./1000;

%% optimized tire contact patch load variation
c_multiplier = 1;

for Ks = [14e3, 22e3]

    F_opt = F_damping*c_multiplier; %output of the 1D-LUT
    file = 'optimized_2DOF';
    sim(file)

    figure(1)
    plot(F_tire.time,F_tire.data,'linewidth',2)
    xlabel('$time$ (s)$','Interpreter','latex','fontSize',16)
    ylabel('$F_{tire}$ (N)$','Interpreter','latex','FontSize',14)
    xticks(0:1:5)
    yticks(-300:50:300)
    ylim([-300,300])
    grid on
    hold on
    set(gca, 'FontSize', 14)
    rms(F_tire.data)
end

legend('optimized', 'non optimized', 'FontSize',14)
```

```

%% optimized vertical acceleration of unsprung masses
c_multiplier = 1.2;

for Ks = [14e3, 22e3]

    F_opt = F_damping*c_multiplier; %output of the 1D-LUT
    file = 'optimized_2DOF';
    sim(file)

    figure(2)
    plot(xs_ddot.time,xs_ddot.data,'linewidth',2)
    xlabel('$time (s)$','Interpreter','latex','fontSize',16)
    ylabel('$\ddot{x}_{s} (m/s^2)$','Interpreter','latex','FontSize',16)
    xticks(0:1:5)
    yticks(-1:0.2:1)
    ylim([-1,1])
    grid on
    hold on
    set(gca, 'FontSize', 14)
    rms(xs_ddot.data)
end

legend('optimized', 'non optimized', 'FontSize',14)

```

Half car Frequency analysis (Bode diagrams) starting from state space - Fig.17

```

clear all
close all
clc

%% Ford Granada half car Parameters
a = 1.25; %m
b = 1.51; %m
ms = 690; %kg
muf = 40.5; %kg
mur = 45.4; %kg
ksf = 17e3; %kg
ksr = 22e3; %kg
ktf = 192e3; %N/m
ktr = 192e3; %N/m
c = 1.5e3; %Ns/m
I = 1222; %kg*m^2

U = 15; %m/s longitudinal speed
tau = (a+b)/U; %s road signal delay at rear axle

%% state space matrices

%x = [xs, xs_dot, theta, theta_dot, xf, xf_dot, xr, xr_dot]' states
%u = [yf, yr]' input

legend_entries = {};

for c = [1, 0.5e3, 1e3, 1.5e3, 2e3, 2.5e3]

    A = [

```

```

0      0      1      0      0
0      0      0      0      0;
      -(ksf+ksr)/ms      -(2*c)/ms      (ksf*a -
ksr*b)/ms      (c*a - c*b)/ms      ksf/ms      c/ms
ksr/ms      c/ms;
0      0      0      0      0
1      0      0      0      0;
      (ksf*a - ksr*b)/I      (c*a - c*b)/I      -(ksf*a^2 +
ksr*b^2)/I      -(c*a^2 + c*b^2)/I      -ksf*a/I      -c*a/I
ksr*b/I      c*b/I;
0      0      0      0      0
0      ksf/muf      0      1      0      0;
-c*a/muf      -(ksf+ktf)/muf      c/muf      -c/muf      -ksf*a/muf
0      0      0      0      0      0;
0      0      0      0      0      0
0      ksr/mur      0      c/mur      0      1;
c*b/mur      0      0      0      0      ksr*b/mur
      -(ksr+ktr)/mur      -c/mur
];

```

```

B = [
0      0;
0      0;
0      0;
0      0;
0      0;
ktf/muf 0;
0      0;
0      ktr/mur
];

```

```
%if states are desired as output
```

```
C = eye(8);
```

```
D = zeros(8,2);
```

```
%sprung mass acceleration bode
```

```
C_acc_s = A(2,:);
```

```
D_acc_s = B(2,:);
```

```
sys_acc_s = ss(A, B, C_acc_s, D_acc_s);
```

```
[mag1,~,w1] = bode(sys_acc_s(:,1)); %input = front road profile
```

```
mag1 = squeeze(mag1);
```

```
f1 = w1 / (2*pi);
```

```
figure(1);
```

```
if c == 1.5e3
```

```
    semilogx(f1, 20*log10(mag1), 'r', LineWidth=2); % plot in Hz
```

```
else
```

```
    semilogx(f1, 20*log10(mag1), LineWidth=1); % plot in Hz
```

```
end
```

```
xlim([1e-1,1e2])
```

```
xlabel('Frequency (Hz)', 'FontSize',16);
```

```
ylabel('Magnitude (dB)', 'FontSize',16);
```

```
set(gca, 'FontSize', 14)
```

```
title('Heave acceleration Bode plot');
```

```
grid on;
```

```
hold on
```

```
legend_entries{end+1} = sprintf('c = %.1f KN/m', c/1000);
```

```
%pitch acceleration bode
```

```
C_acc_pitch = A(4,:);
```



```

D_acc_pitch = B(4,:);
sys_acc_pitch = ss(A, B, C_acc_pitch, D_acc_pitch);
[mag2,~,w2] = bode(sys_acc_pitch(:,1)); %input = front road profile
mag2 = squeeze(mag2);
f2 = w2 / (2*pi);
figure(2);
if c == 1.5e3
    semilogx(f2, 20*log10(mag2), 'r', LineWidth=2); % plot in Hz
else
    semilogx(f2, 20*log10(mag2), LineWidth=1); % plot in Hz
end
xlim([1e-1,1e2])
xlabel('Frequency (Hz)','FontSize',16);
ylabel('Magnitude (dB)','FontSize',16);
set(gca, 'FontSize', 14)
title('Pitch acceleration Bode plot');
grid on;
hold on
end

```

```

legend(legend_entries, 'Location', 'best', FontSize=10);

```

Half car optimized and non-optimized force and acceleration comparison - Fig.19 and Table 14

```

%% Parameters
a = 1.25; %m
b = 1.51; %m
ms = 690; %kg
muf = 40.5; %kg
mur = 45.4; %kg
Ksf = [14e3, 17e3]; %kg    optimized and non-optimized values
Ksr = [14e3, 22e3]; %kg    optimized and non-optimized values
c_multiplier = [1.1, 1];  optimized and non-optimized values
ktf = 192e3; %N/m
ktr = 192e3; %N/m
I = 1222; %kg*m^2

%1-D LUT for Low damping damper-dyno curve
data = readmatrix("Damper_Low.txt");
v_relative = data(:,1)/1000; %m/s
F_damping = data(:,2); %N

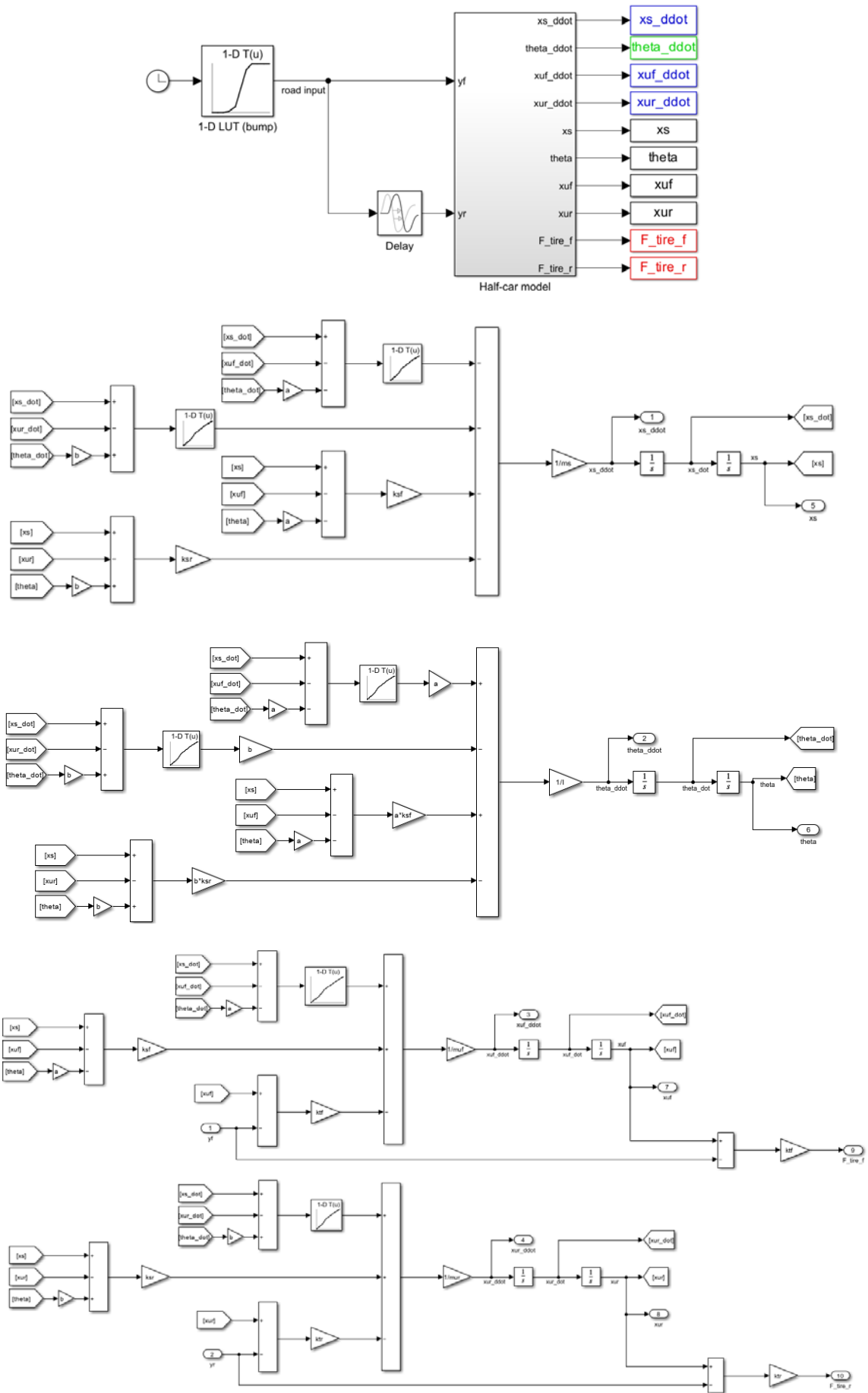
speed = 15; %m/s longitudinal speed
tau = (a+b)/speed; %s road signal delay at rear axle
T_sim = 5; % s

% Bump input (accounting for 0.3s transient)
bump = readmatrix('step', 'NumHeaderLines', 2); %imported from Adams
t = bump(3:end,2);
y = bump(3:end,1)./1000;

for i = 1:length(c_multiplier)
    ksf = Ksf(i); %N/m
    ksr = Ksr(i); %N/m
    F_opt = c_multiplier(i)*F_damping; %N

    file = 'HalfCar_optimized';
    sim(file)
end

```



```

figure(1)
plot(xs_ddot.time,xs_ddot.data,'linewidth',2)
xlabel('time (s)','Interpreter','latex','FontSize',16)
ylabel('$\ddot{x}_s$ (m/s^2)','Interpreter','latex','FontSize',16)
xticks(0:1:5)
yticks(-0.8:0.2:0.8)
ylim([-0.8,0.8])
grid on
hold on
fprintf('xs_ddot = %.4f ', rms(xs_ddot.data));
set(gca, 'FontSize', 14)
legend('optimized', 'non optimized')

figure(2)
plot(theta_ddot.time, theta_ddot.data, 'linewidth',2)
xlabel('time (s)','Interpreter','latex','FontSize',16)
ylabel('$\ddot{\theta}$ (rad/s^2)','Interpreter','latex','FontSize',16)
grid on
hold on
fprintf('theta_ddot = %.4f ', rms(theta_ddot.data));
set(gca, 'FontSize', 14)
legend('optimized', 'non optimized')

figure(3)
plot(F_tire_f.time,F_tire_f.data, 'linewidth',1.5)
xlabel('time (s)','Interpreter','latex','FontSize',16)
ylabel('$F_{tire,f}$ (N)','Interpreter','latex','FontSize',16)
xticks(0:1:5)
yticks(-300:150:300)
ylim([-300,300])
grid on
hold on
fprintf('Front tire force = %.4f ', rms(F_tire_f.data));
set(gca, 'FontSize', 14)

figure(4)
plot(F_tire_r.time,F_tire_r.data, 'linewidth',1.5)
xlabel('time (s)','Interpreter','latex','FontSize',16)
ylabel('$F_{tire,r}$ (N)','Interpreter','latex','FontSize',16)
xticks(0:1:5)
yticks(-300:150:300)
ylim([-300,300])
grid on
hold on
fprintf('Rear tire force = %.4f ', rms(F_tire_r.data));
set(gca, 'FontSize', 14)
legend('optimized', 'non optimized')
end

```